Placenta

By

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TW Sadler. Third month to birth: In Langman's medical embryology, 15th edition,2024, Chapter 8: P108-128.

Learning objectives

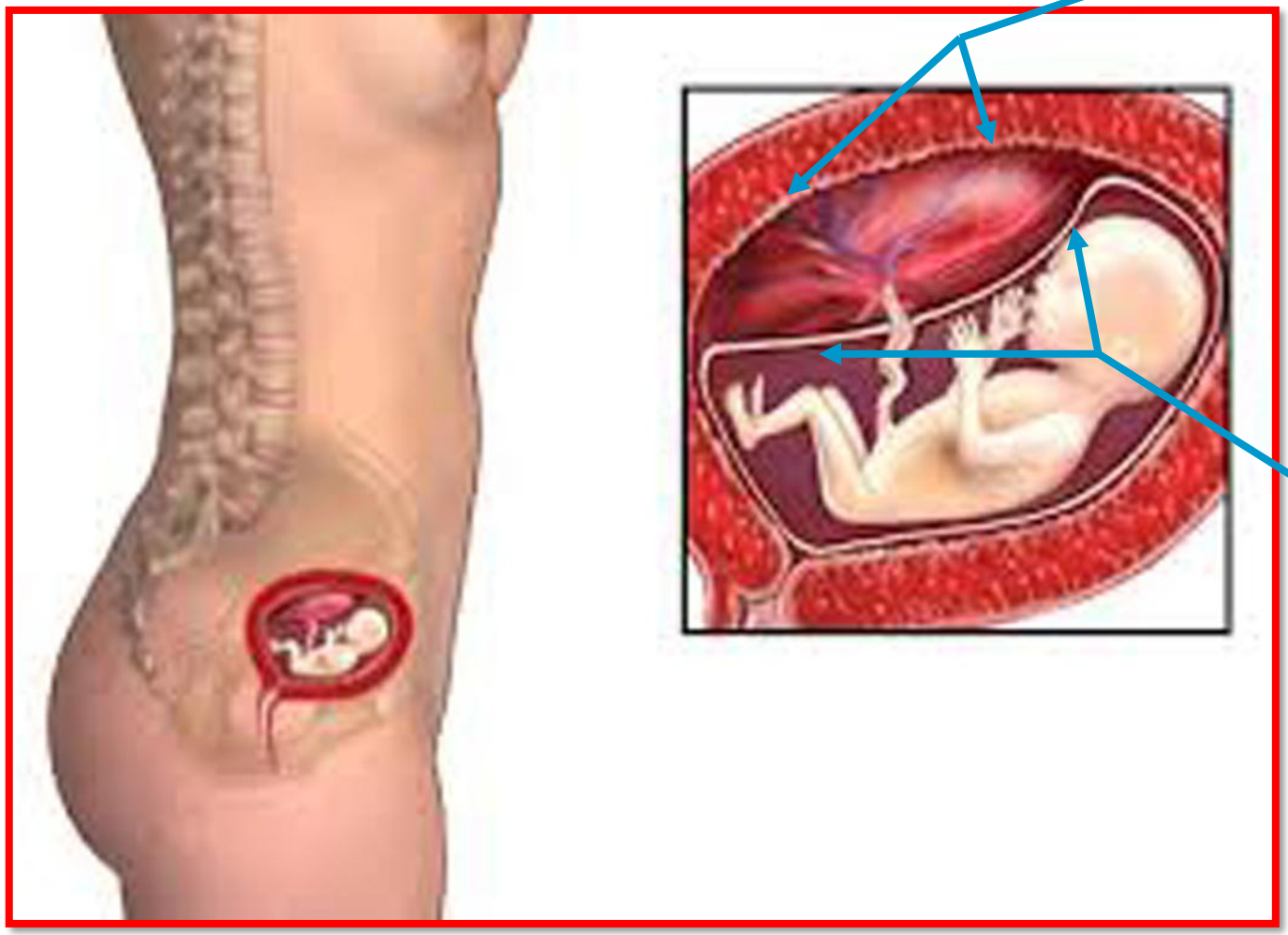
- Gross anatomy of placenta.
- Parts and structure of placenta.
- Placental circulation.
- Development of placenta.
- Placental barrier.
- Functions of placenta.
- Clinical notes.
- Anomalies of placenta.

Gross Anatomy

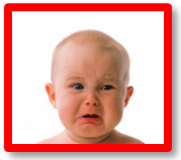
- The full term placenta has the following features:
  - **Shape:**
    - Discoid shape.
  - **Diameter:**
    - 15-25 cm.
  - **Thickness:**
    - About 3 cm
  - **Weight:**
    - 500-600 gm.

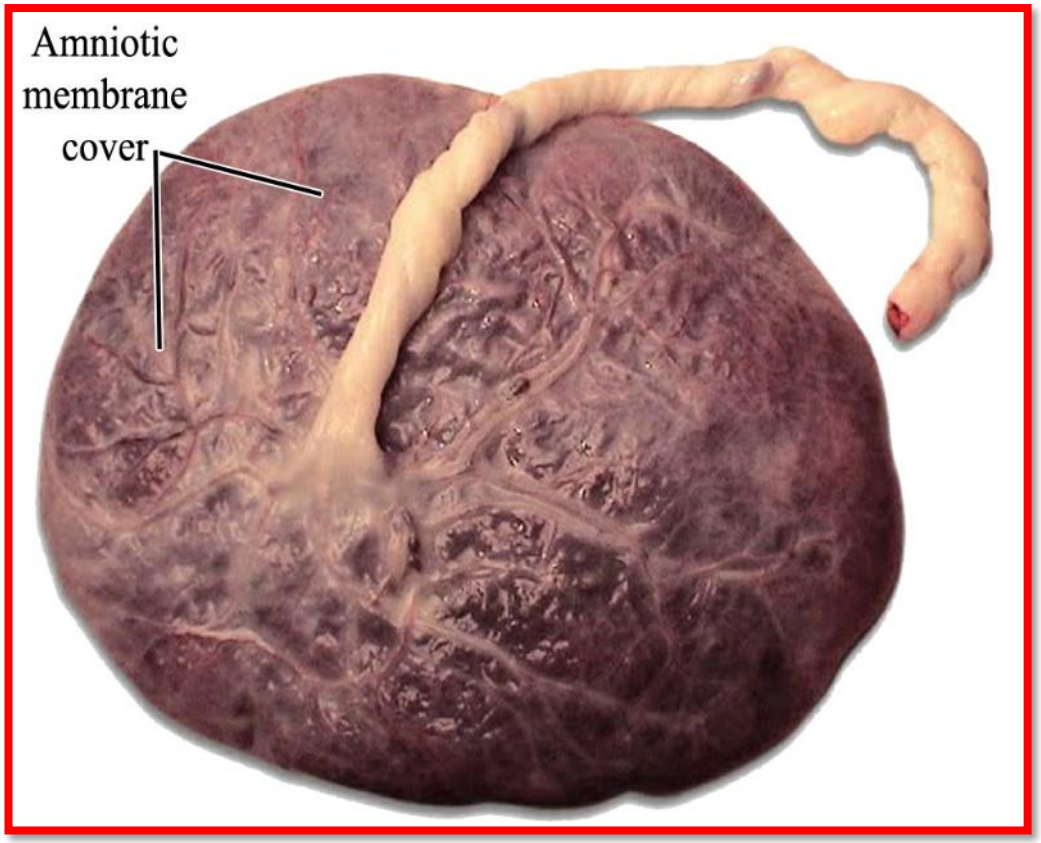


➢ Surfaces: (two surfaces) 2



1. Fetal surface



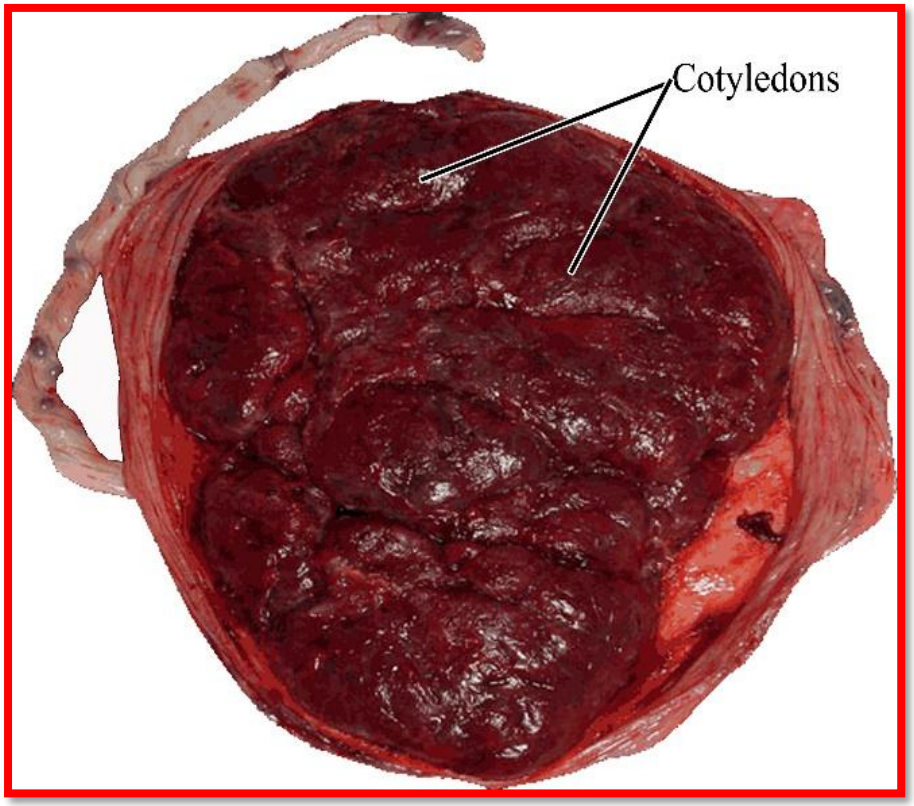


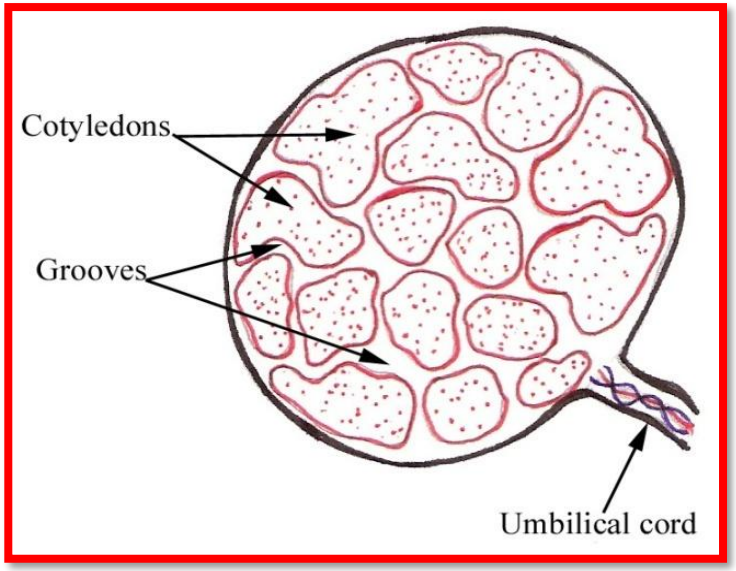


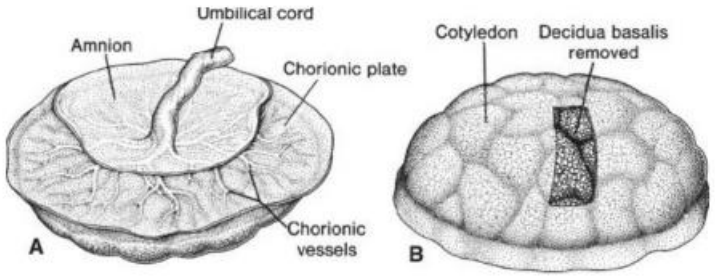
- It is smooth and covered by the **amnion.**
- It has **chorionic vessels** converging towards the umbilical cord.
- The **umbilical cord** is attached to this surface **(eccentric).**

2. Maternal surface



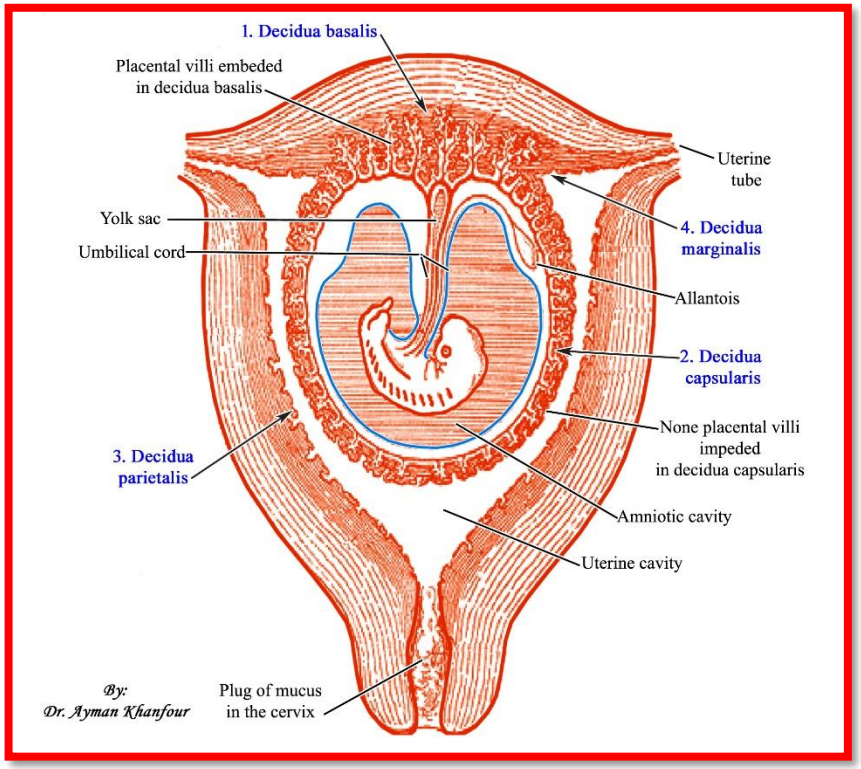


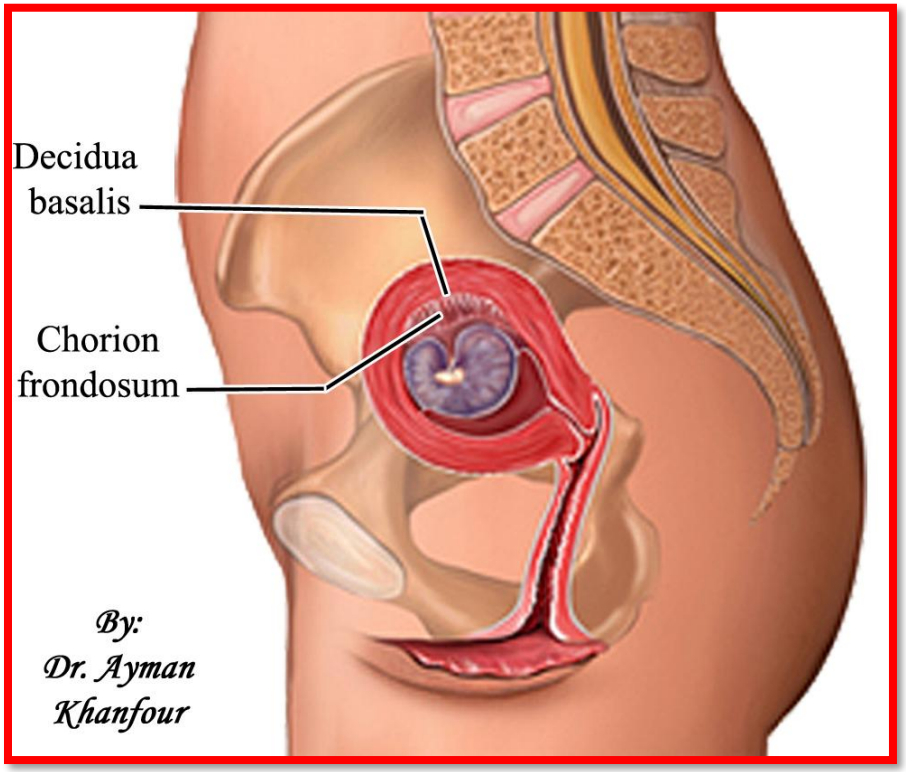




- It is irregular and contains **15-20 cotyledons** with deep grooves in between made by the **placental septa.**

Structure of the Placenta





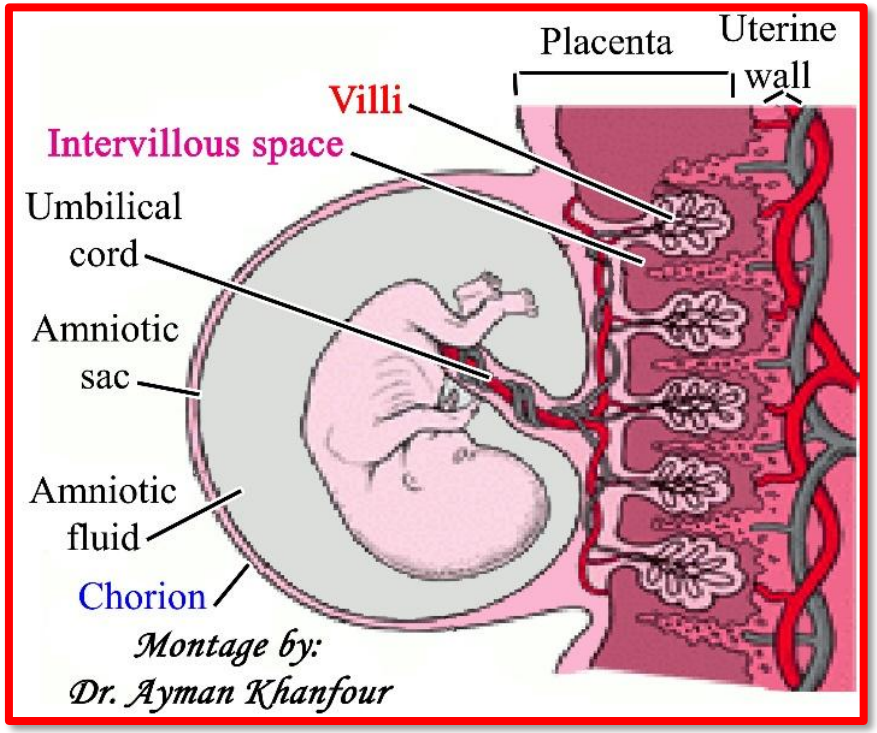
- **The placenta has two components:**

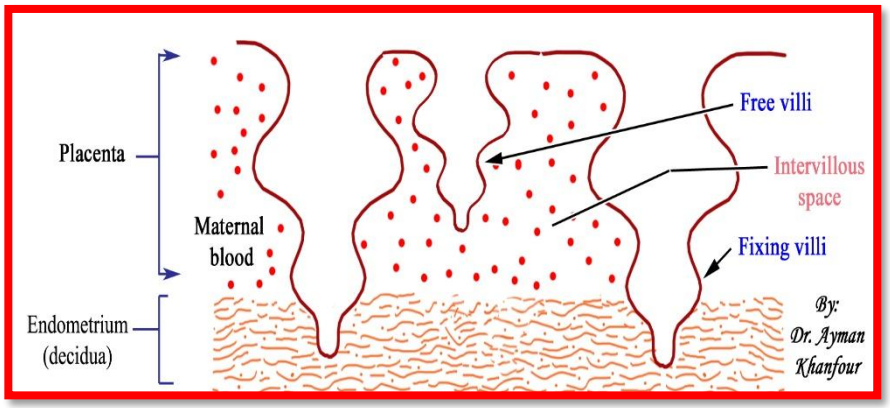
  a. Fetal part (chorion frondosum)

  b. Maternal part (decidua basalis)

1. Fetal part







- It is formed by the ***chorion frondosum.***

- These are the chorionic villi opposite the **decidua basalis.**

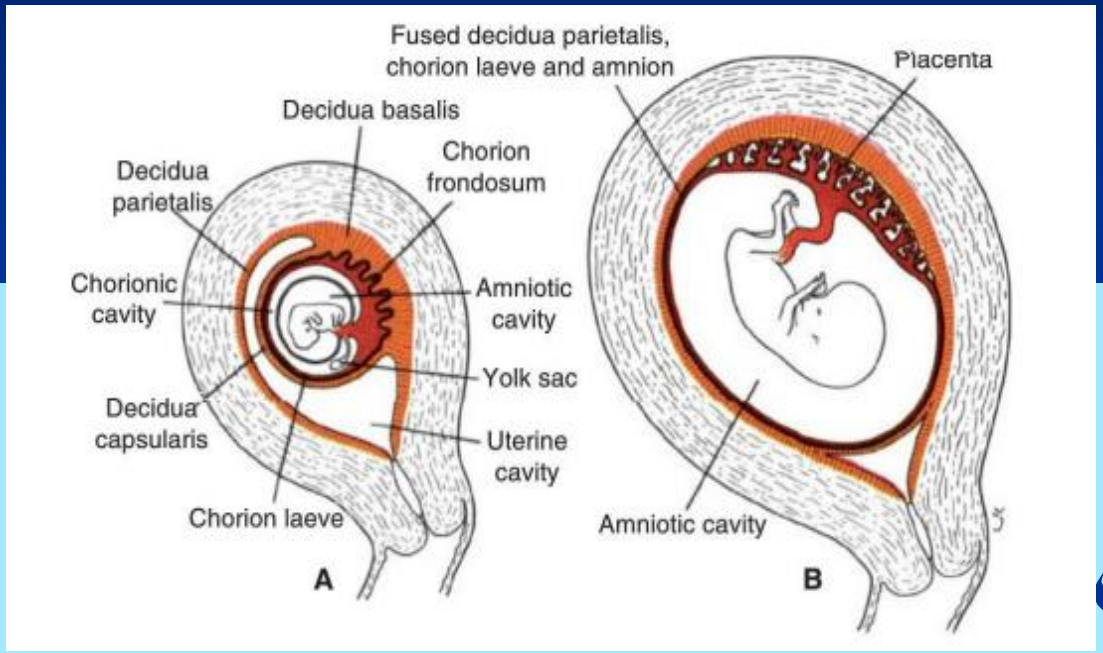
Types:

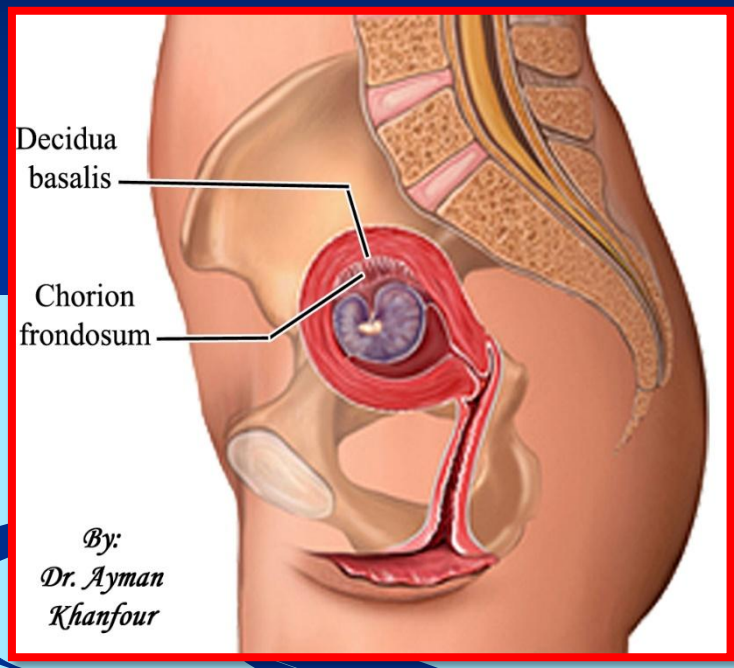
a. Fixing (anchoring) villi which penetrates through decidua basalis.

b. Free (nourishing) villi which don't penetrate the decidua basalis.

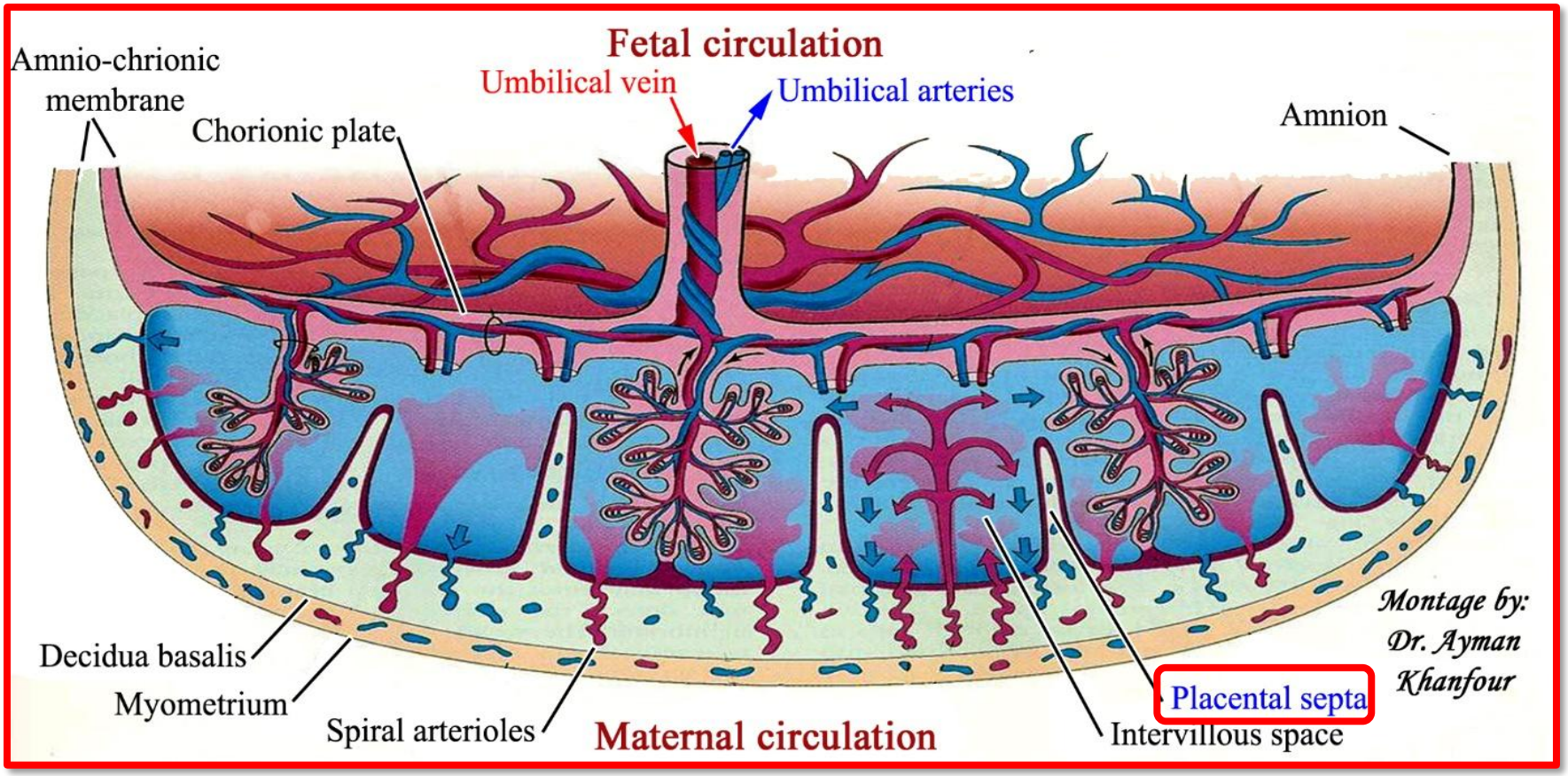
2. Maternal part







- It is formed by the *decidua basalis.*



- Placenta is formed of **15-20 cotyledons** with deep grooves in between made by **placental septa.**
- **Placental septa** are incomplete decidual septa and project into intervillous spaces.

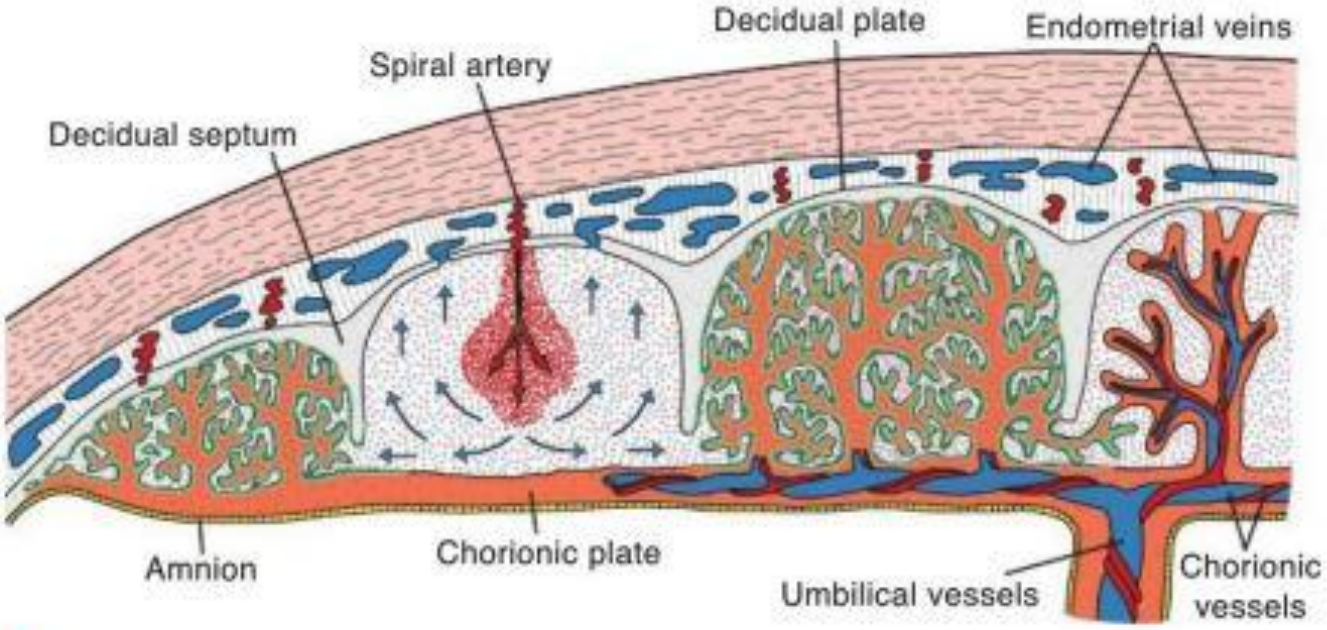
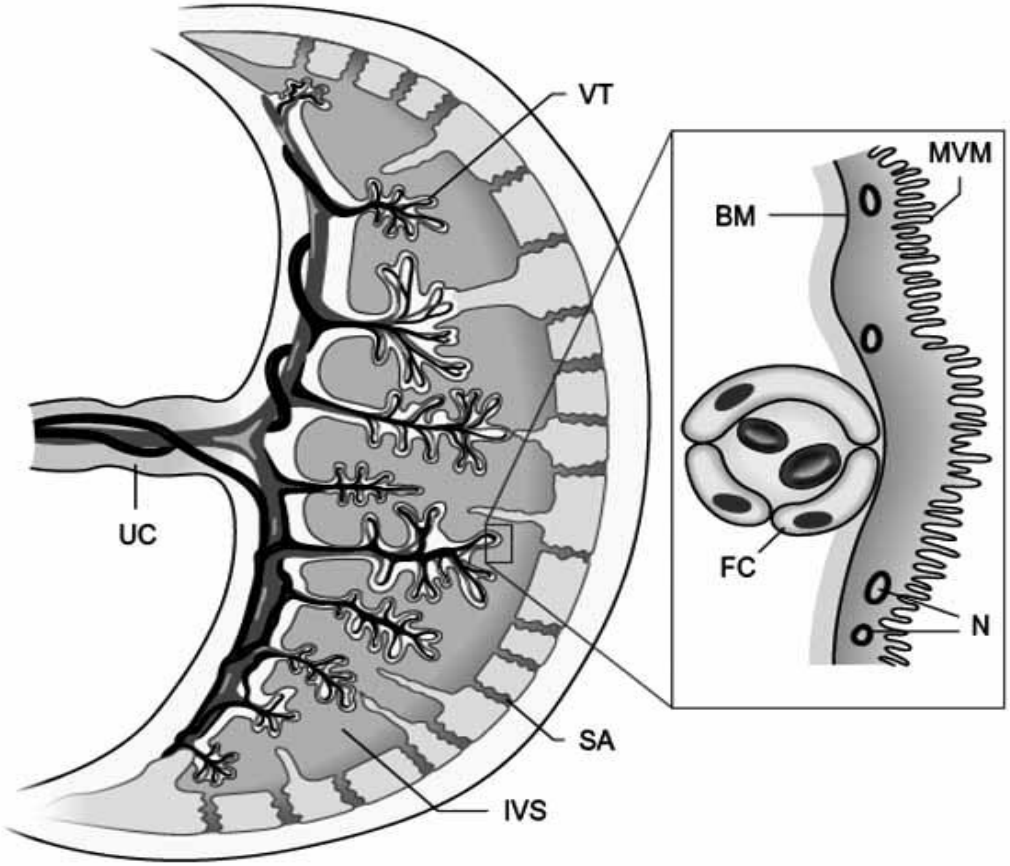


FIGURE 8.13 The placenta in the second half of pregnancy. The cotyledons are partially separated by the decidual [maternal] septa. Most of the intervillous blood returns to the maternal circulation by way of the endometrial veins. A small portion enters neighboring cotyledons. The intervillous spaces are lined by syncytium.

- Spaces between the villi form the **intervillous spaces.**
- The **intervillous spaces** are **communicating** with each other.
- They are filled with **maternal blood** leaked from **80-100 spiral arteries** eroded by chorionic villi.
- Intervillous spaces of mature placenta contains 150 ml of blood.



- As pregnancy advances, the placenta increases in:

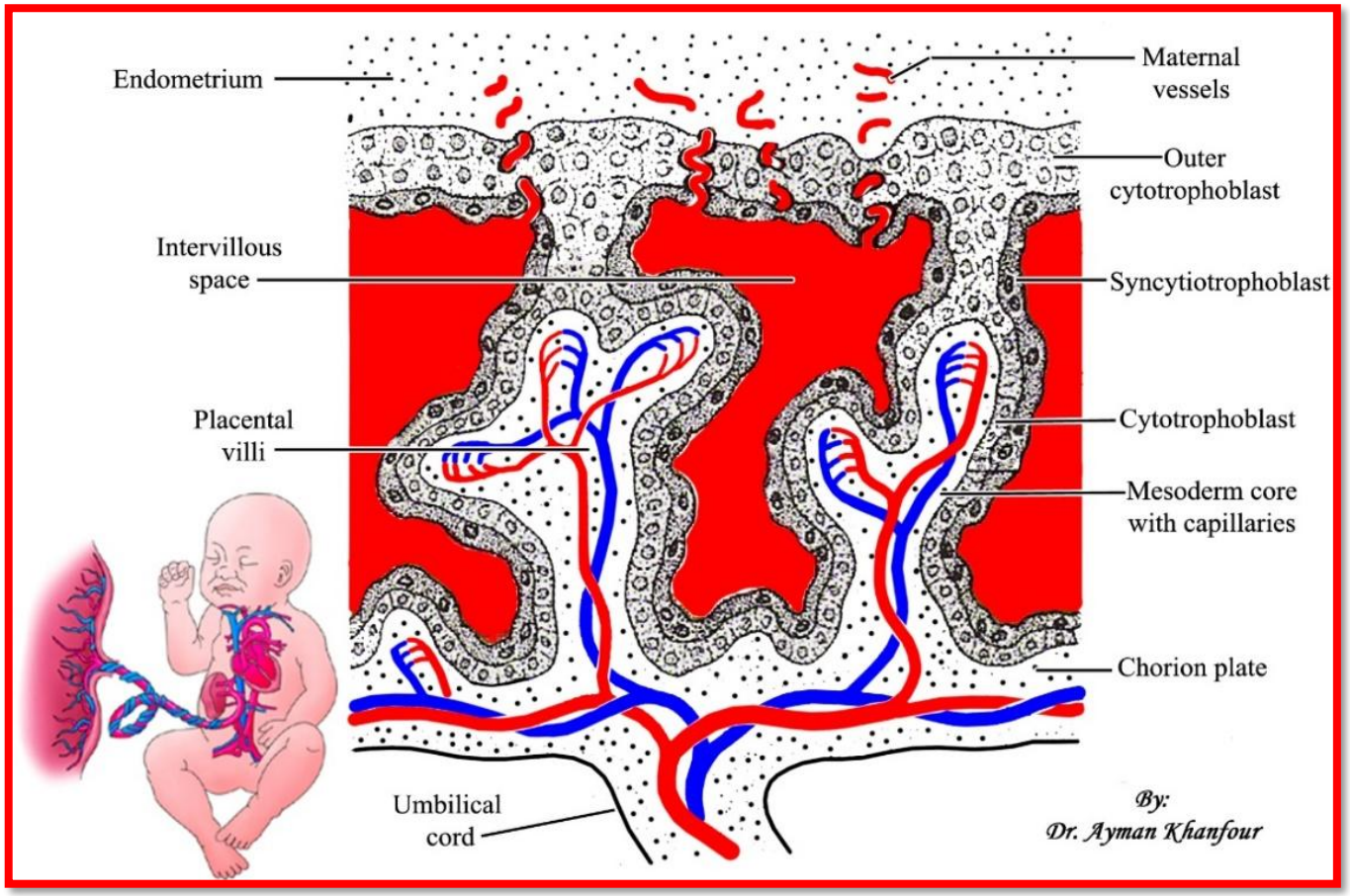
  a. **Size** by formation of **new villi.**

  b. **Thickness** due to the **extensive arborization** (branching) of the villi.

Nourishing villi have brush border consisting of microvilli increasing the surface area to 4-14 m^2.

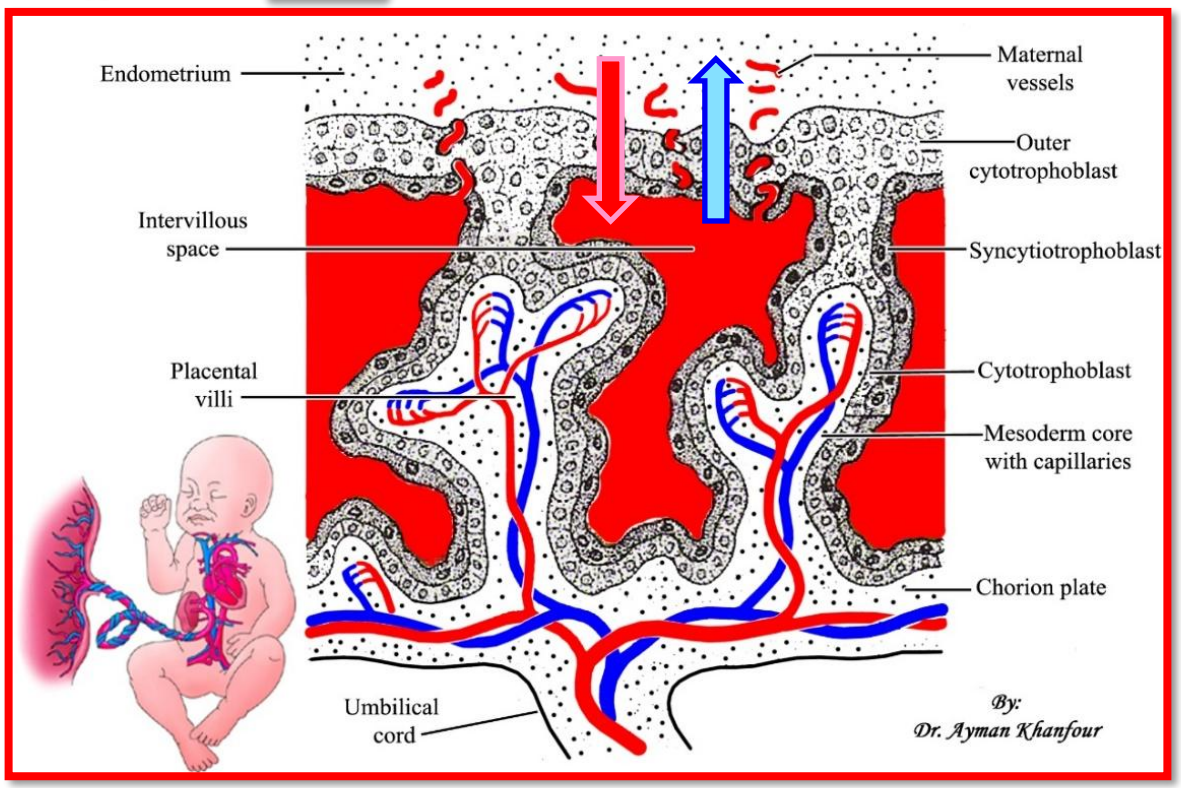
Placental Circulation

{Maternal part}



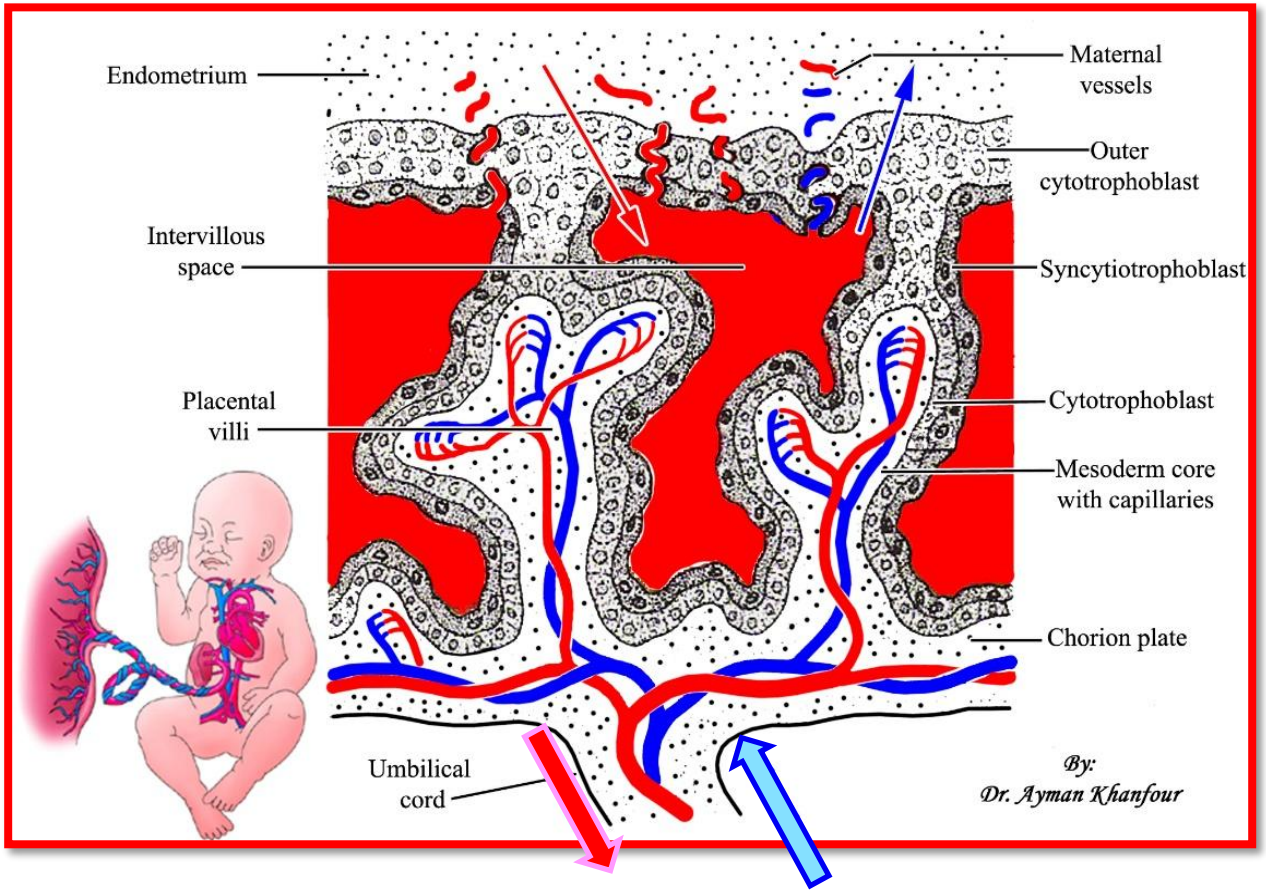
{Fetal part}

1. Maternal part



- Chorionic villi pierce the spiral endometrial arterioles in the decidual plate.
- **Arterial oxygenated blood** from spiral endometrial arterioles (higher pressure ↑) reaches the intervillous spaces.
- This carries nutrients + antibodies into the intervillous spaces .
- **Deoxygenated blood** passes back towards endometrial venules (lower pressure ↓) in the decidua to the maternal circulation, carrying the fetal Co_2 and waste products.

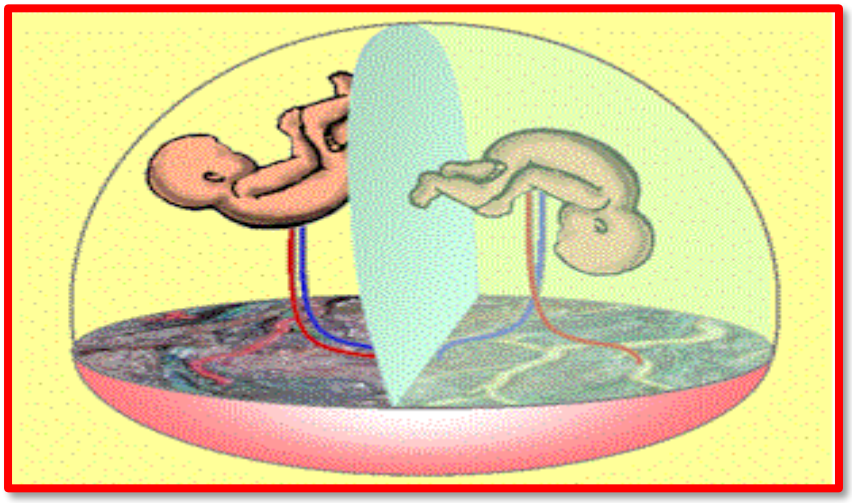
2. Fetal part



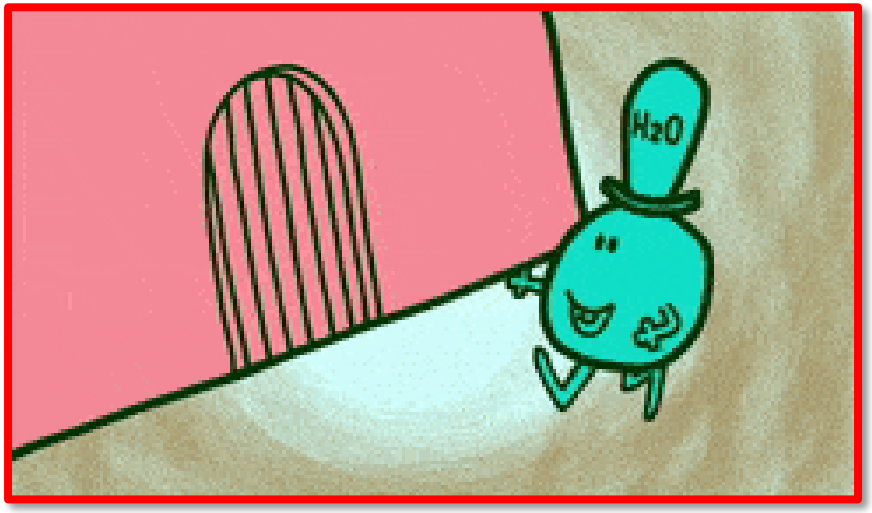
- Oxygen and nutrients pass through the **umbilical vein** to the fetus.
- Fetal CO_2 and waste products pass through the **umbilical arteries** to the mother.
- Exchange of gases + metabolites occurs through the wall of the villi (Placental barrier).

Placental Membrane (Barrier)

- It is a special membrane which separates the maternal and fetal blood.

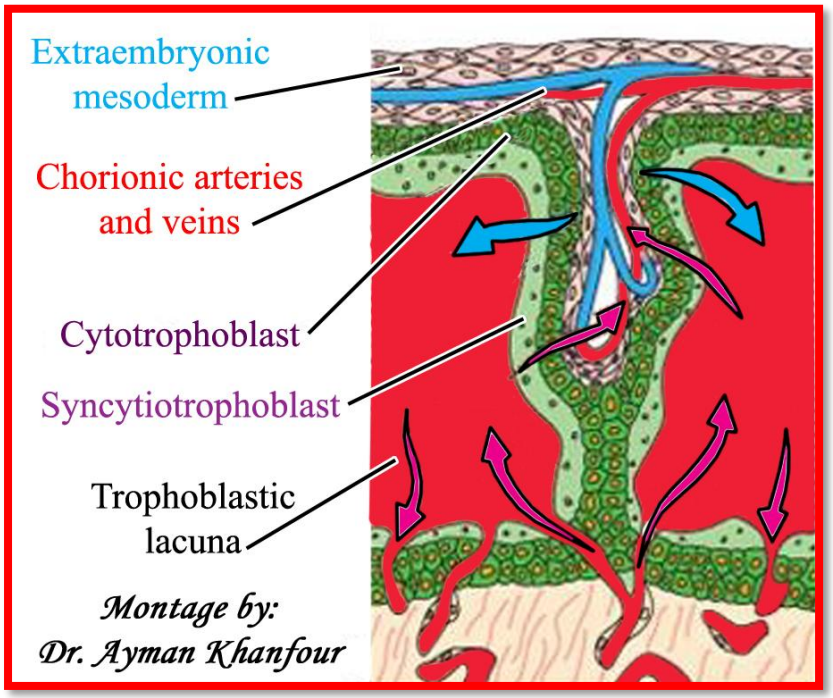


- It has a *selective permeability* which allows passage of certain substances and prevents others.

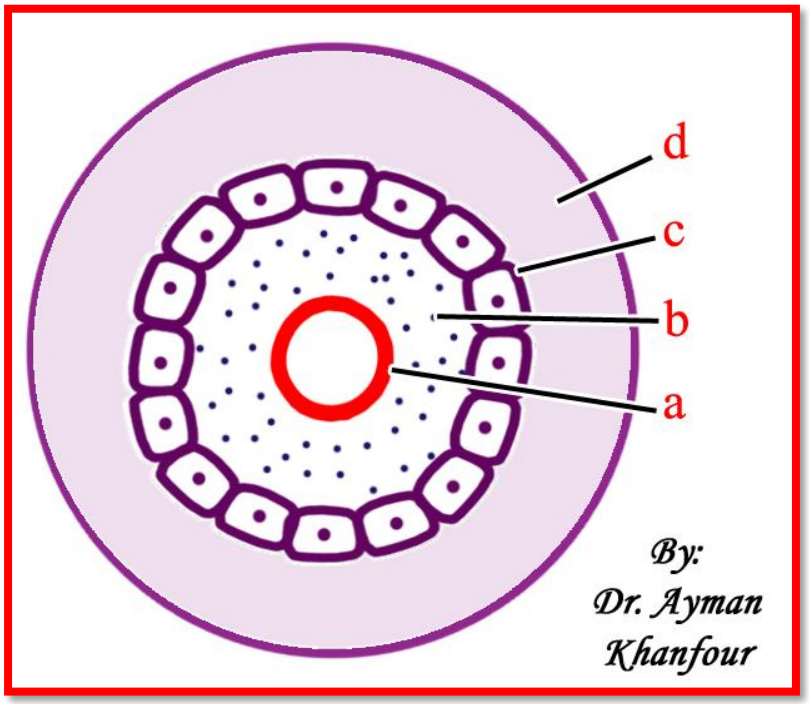


Structure of the placental barrier

A. At early pregnancy, formed of:



(Tertiary villi)



(Placental barrier at early pregnancy)

a. Endothelial lining the fetal vessels.

b. Primary mesoderm.

c. Cytotrophoblast.

d. Syncytiotrophoblast.

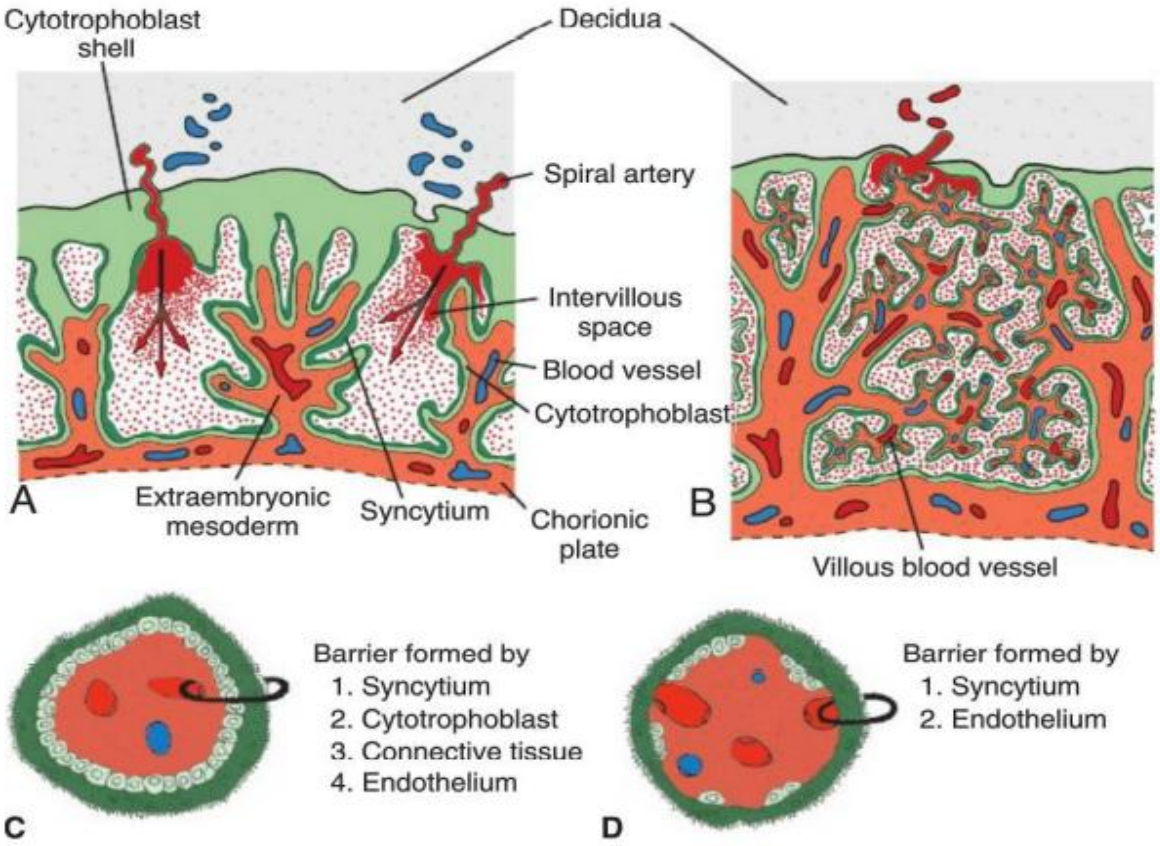
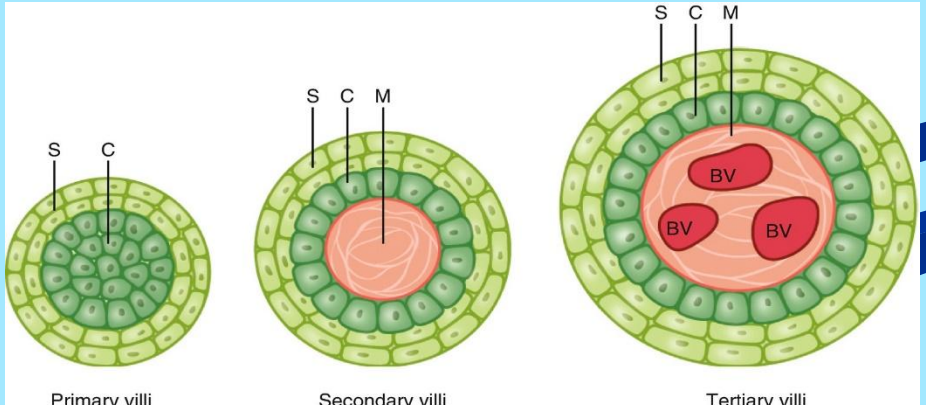
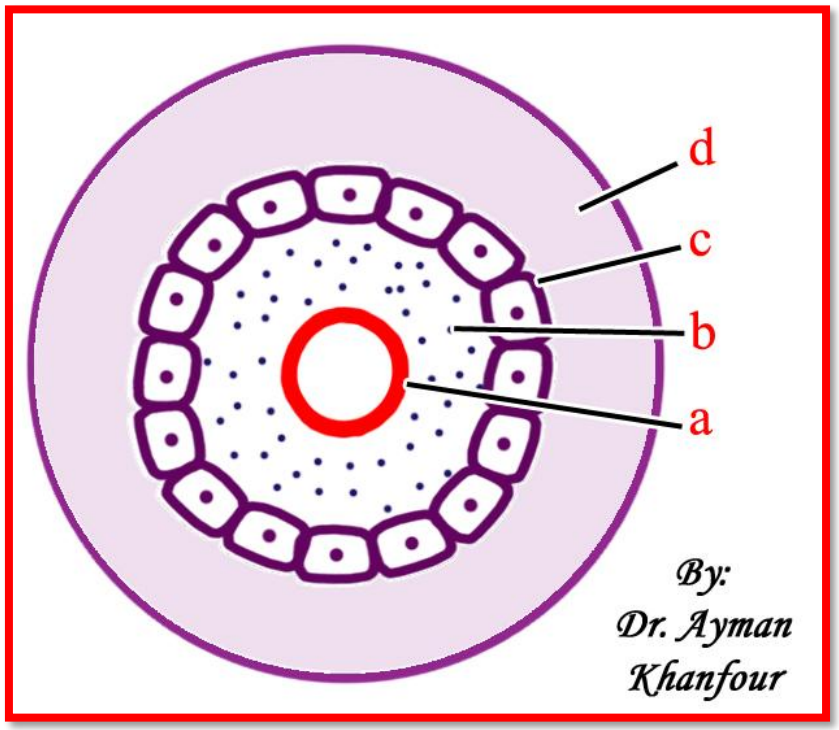


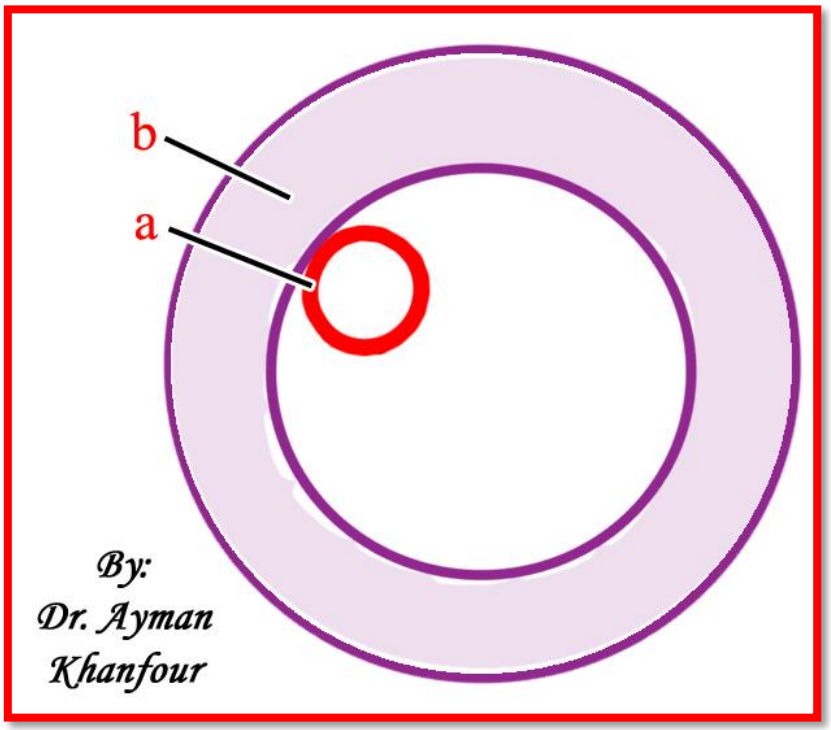
FIGURE 8.8 Structure of villi at various stages of development. **A.** During the fourth week. The extraembryonic mesoderm penetrates the stem villi in the direction of the decidual plate. **B.** During the fourth month. In many small villi, the wall of the capillaries is in direct contact with the syncytium. **C,D.** Enlargement of the villus as shown in Figures 8.8*A*,*B*.



B. At middle of pregnancy (20 w), formed of:



(Placental barrier at early pregnancy)

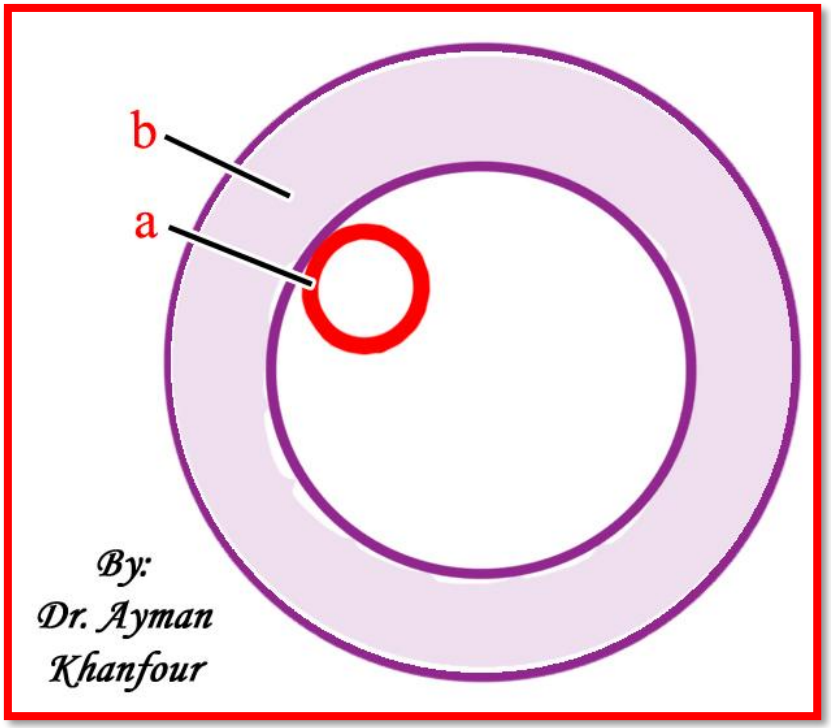


(Placental barrier at middle of pregnancy)

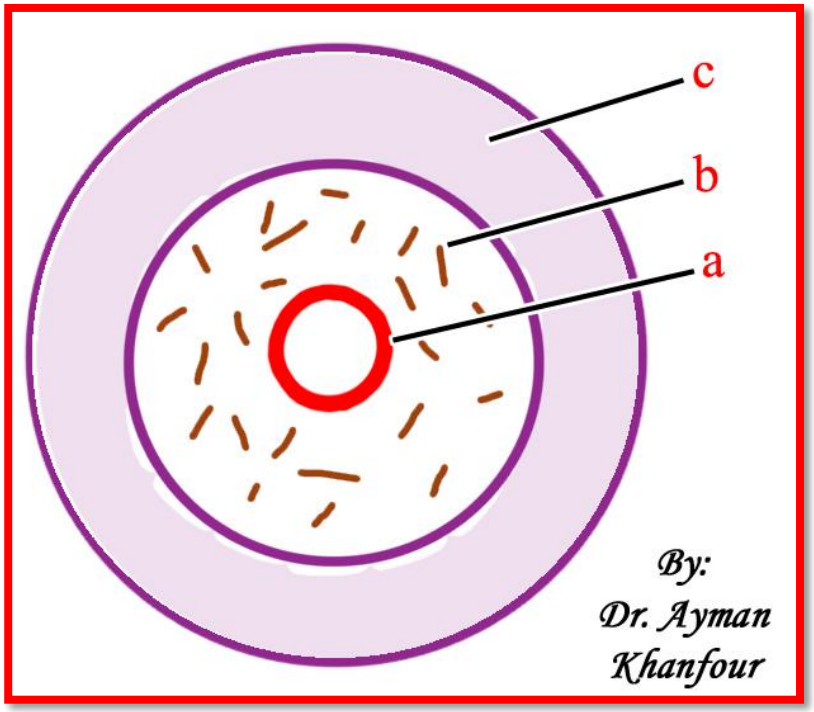
a. Endothelial lining the fetal vessels.

b. Syncytiotrophoblast.

C. At the end of pregnancy, formed of:



(Placental barrier at middle of pregnancy)



(Placental barrier at end of pregnancy)

a. Endothelial lining the fetal vessels.

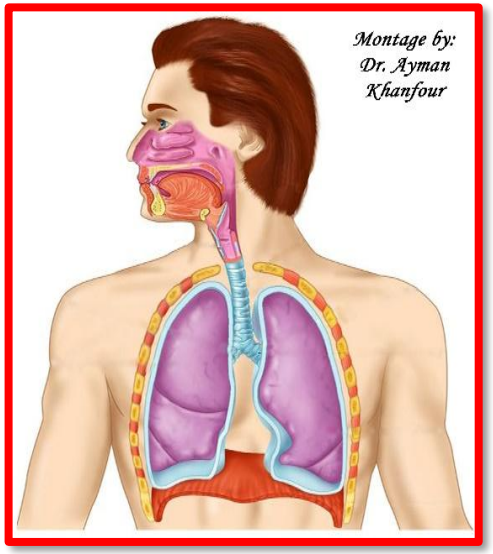
b. Fibrinoid material which surround the fetal blood vessels.

c. Syncytiotrophoblast.

Functions of Placenta

1. Respiration

- It allows for exchange of gases where oxygen, carbon dioxide are transported by simple diffusion.



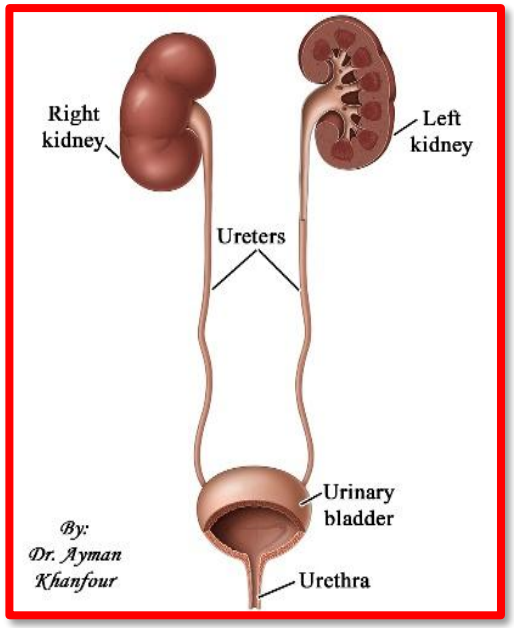
2. Nutrition

- It allows for passage of electrolytes and nutrients as amino acids, fatty acids, carbohydrates and vitamins.



3. Excretion

- It replaces the kidney function to excrete waste products.



4. Protection

- It transmits maternal antibodies which results in passive immunity.



5. Hormone production

- In the first 2 month, Syncytiotrophoblast secretes:

  . Human chorionic gonadotrophins (HCG).

- By the end of the 4th month, placenta secretes the followings:

  . Progesterone and estrogen (estriol) hormones.

  . Somatomammotropin.

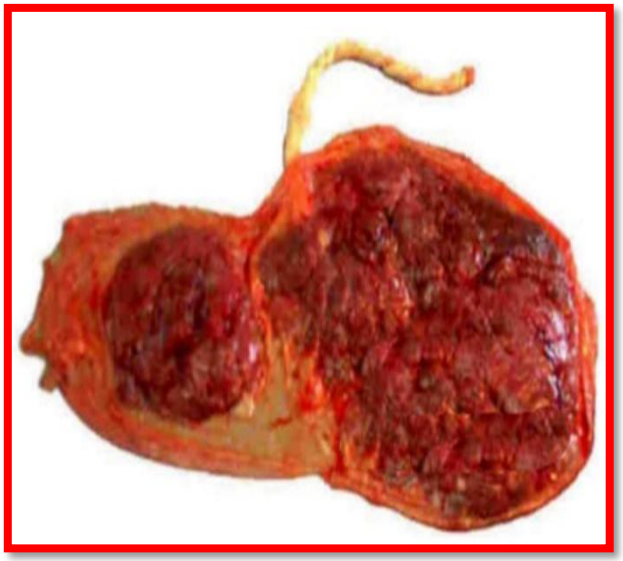
  . Relaxin hormone to soften the ligaments of the pelvis.



Congenital Anomalies

I. According to the number:



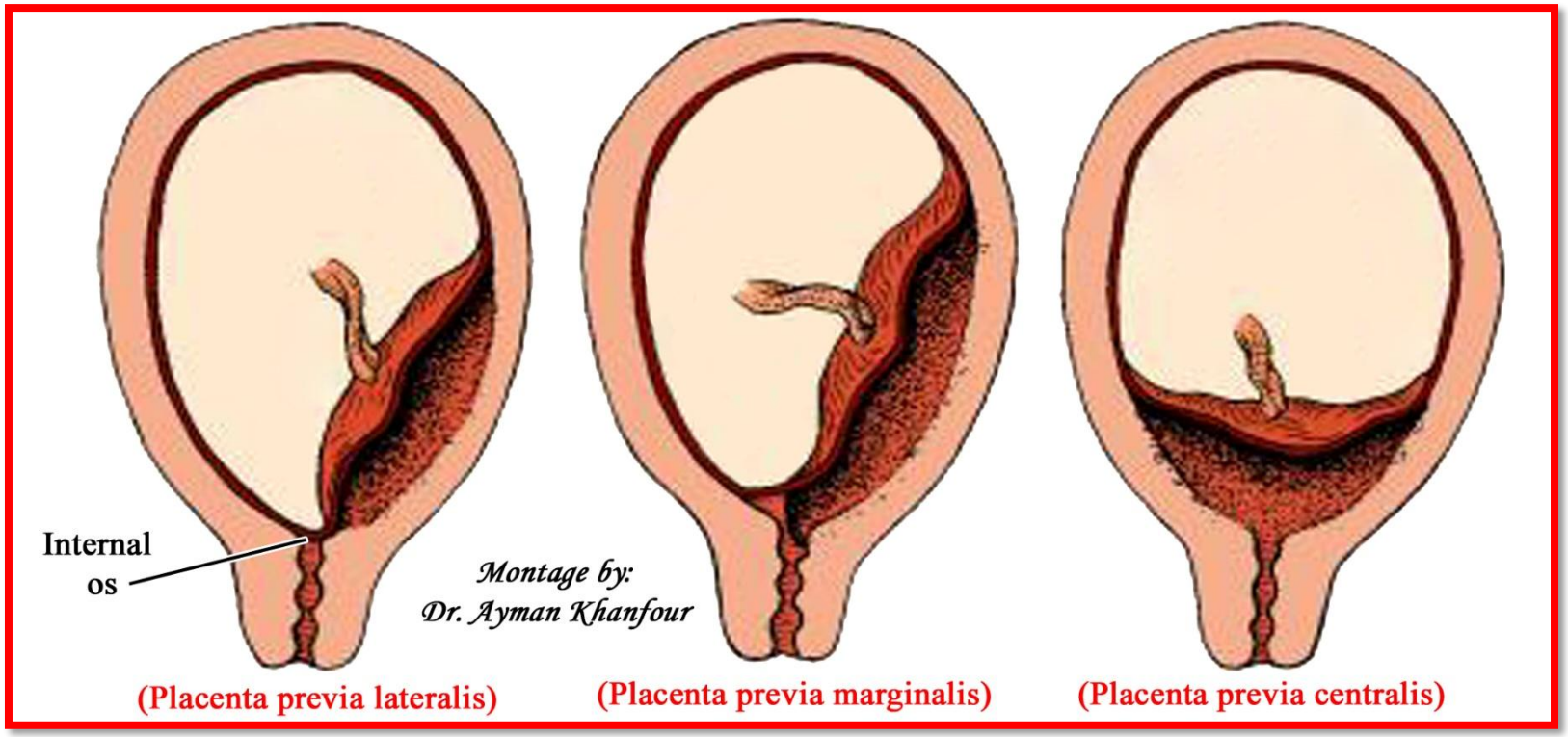


a. Double placenta.

b. Triple placenta.

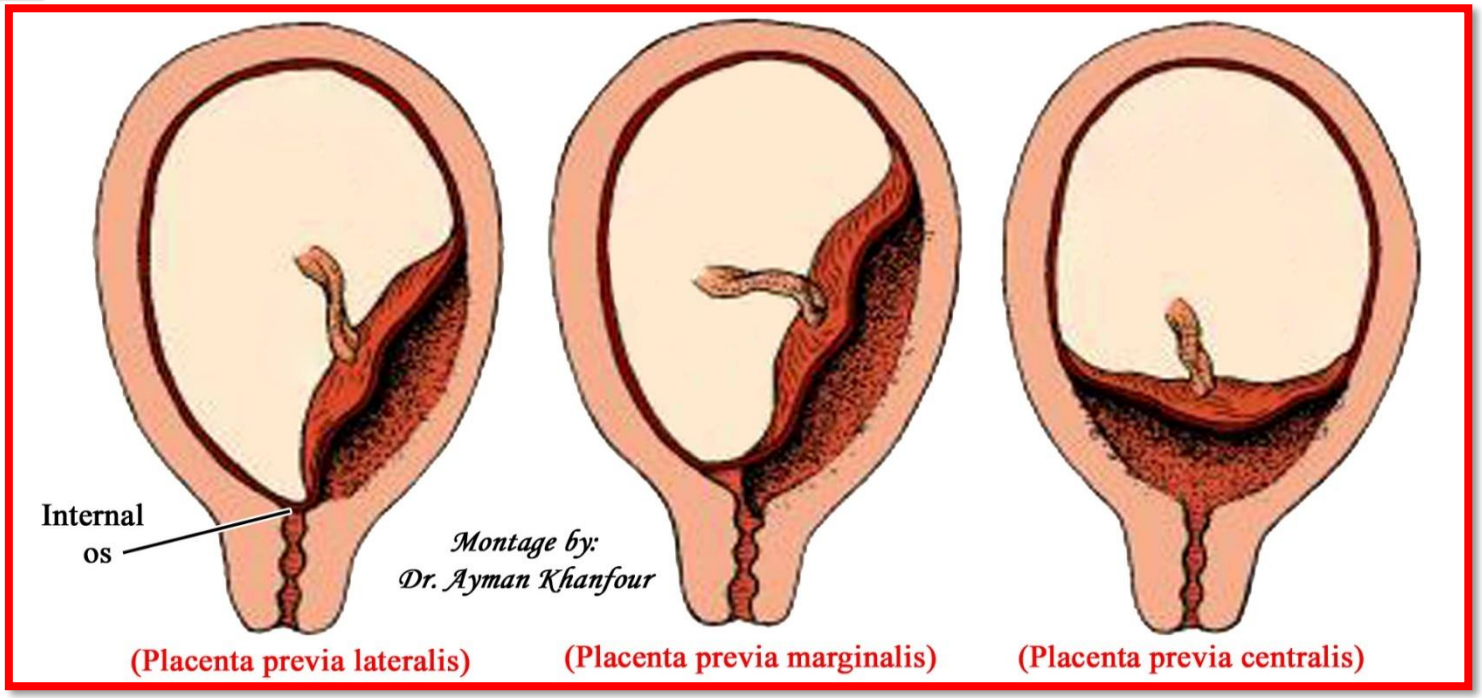
b. Accessory placenta (succenturiate).

II. According to the site:



- Placenta attached to the lower uterine segment is called placenta previa.
- It may cause severe antepartum hemorrhage.
- Placenta previa is usually diagnosed by ultrasonography during pregnancy and delivery must be by Cesarean section to avoid severe antepartum hemorrhage.

Types:



a. Placenta previa lateralis

- It is attached to the lower segment away from the internal os.

b. Placenta previa marginalis

- It covers the internal os incompletely.

c. Placenta previa centralis

- It covers the internal os completely.

III. According to the siz:

a. Small placenta



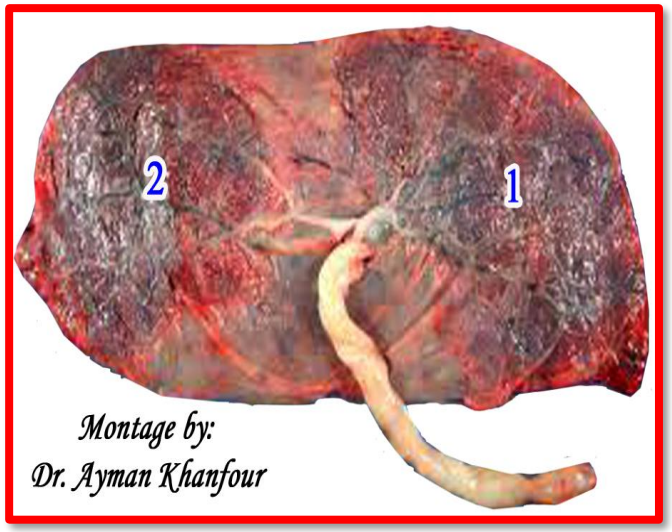
b. Large placenta



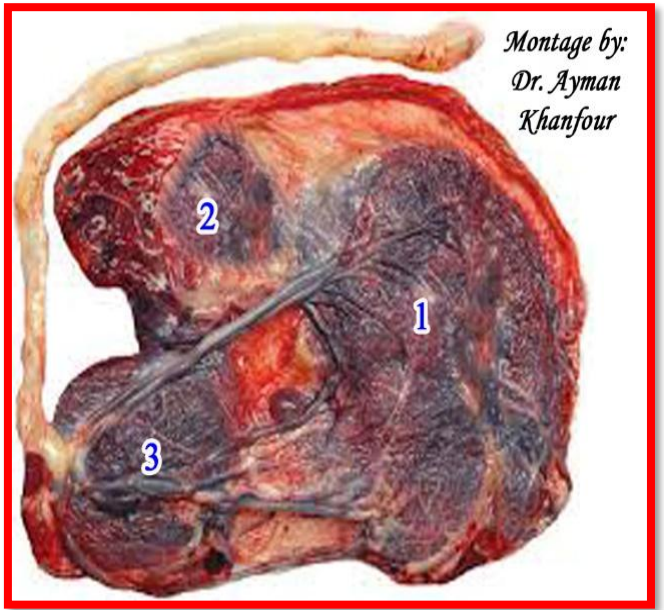
IV. According to the shape:

a. Bilobed placenta



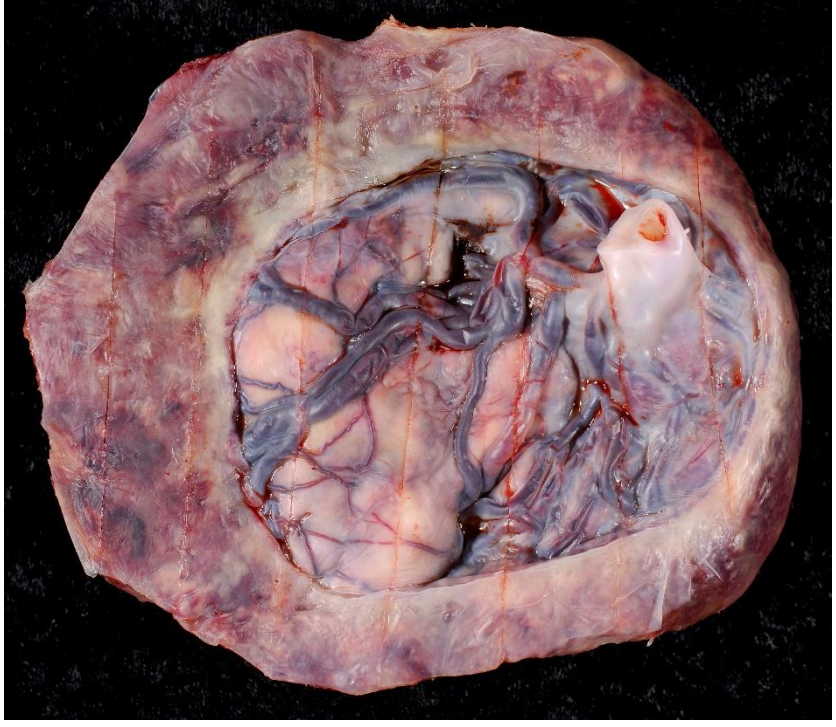


b. Trilobed placenta



c. Circumvallate placenta

- Placenta has a thick ring of membranes (Double fold of chorion and amnion) on its fetal surface.
- Chorionic plate is smaller than decidual plate.

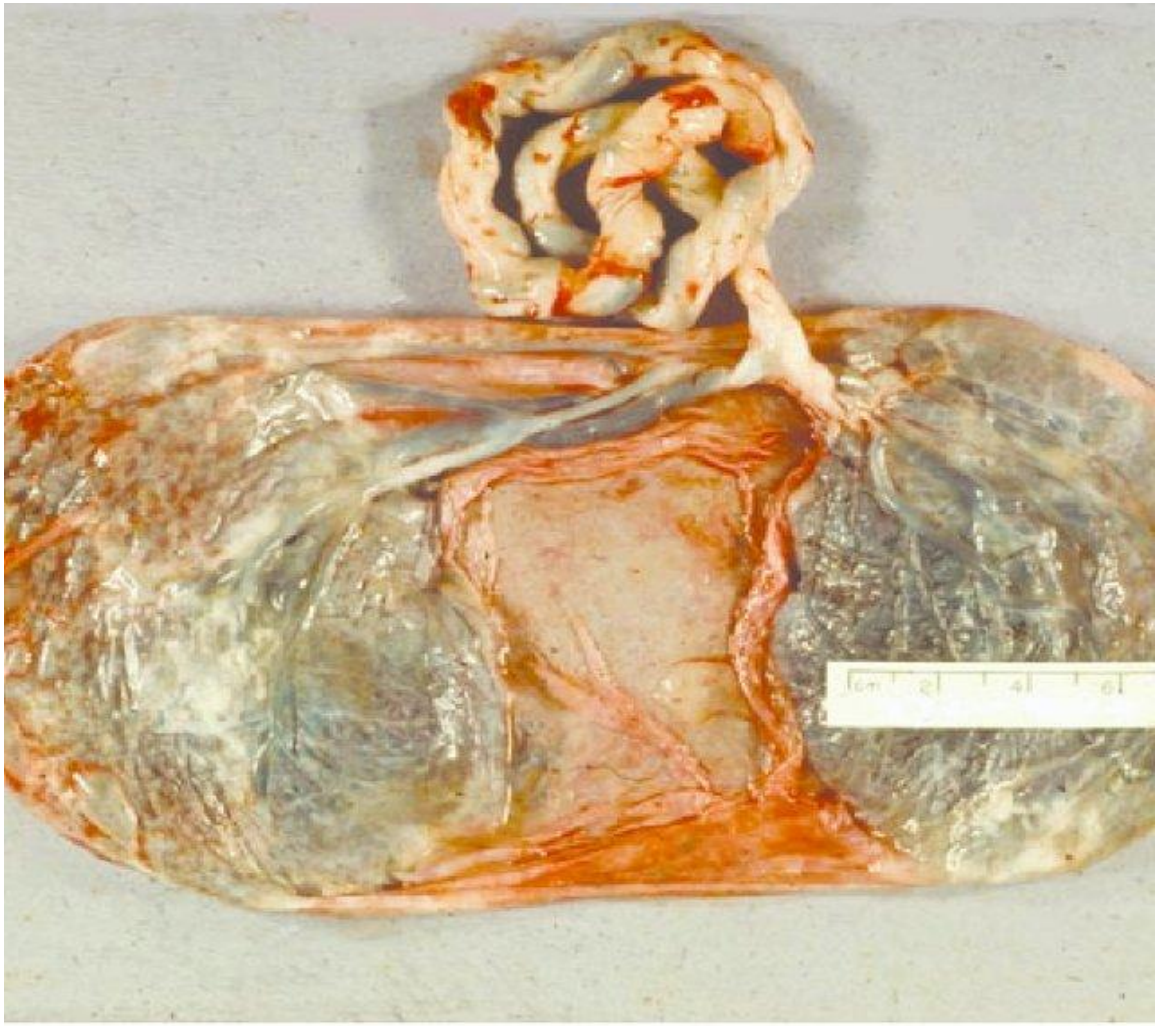


d. Fenestrated placenta

- It contains a central defect.

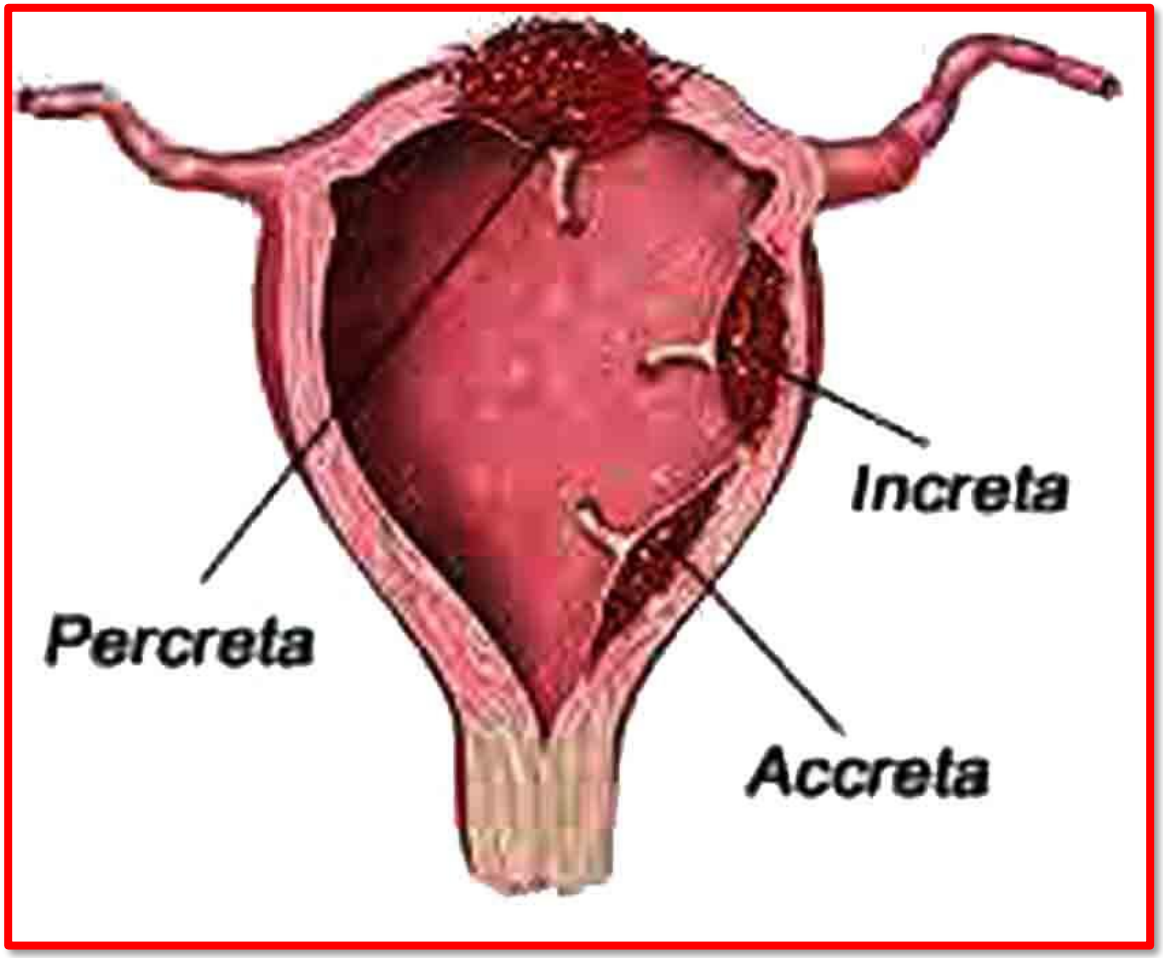


V. According to the thickness:



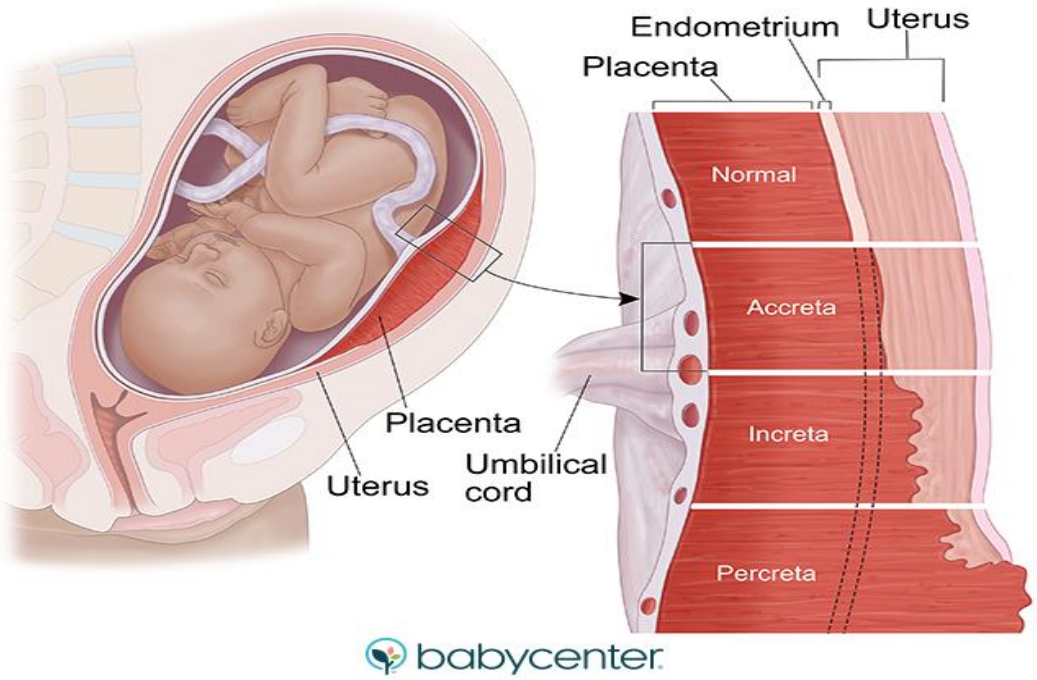
- Membranous (or diffuse placenta).

VI. According to the degree of penetration to the uterine wall:



- Penetration of the chorionic villi extends beyond the endometrium.

➢ Types:



a. Placenta accreta:

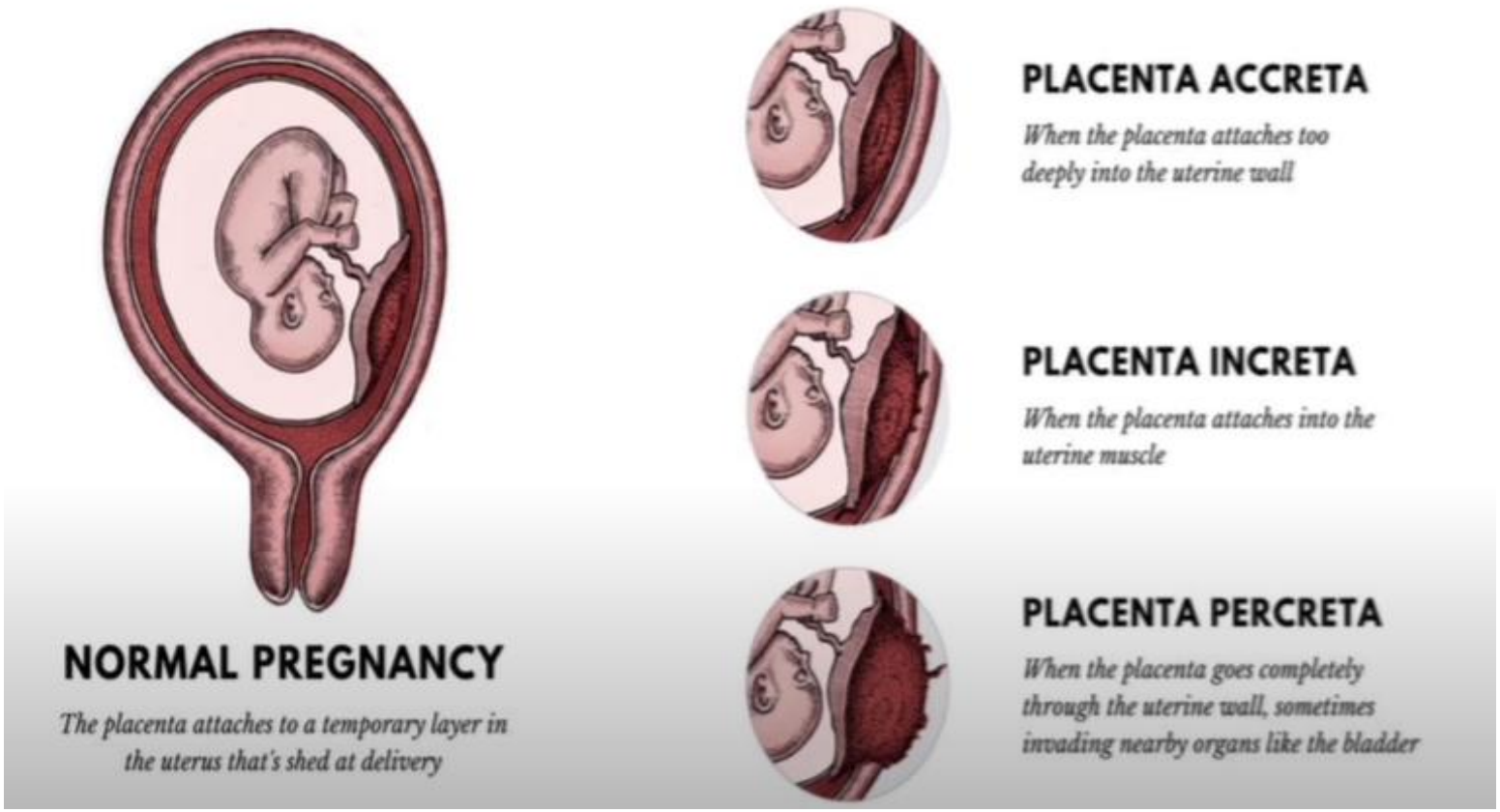
- Excessive penetration of the endometrium.

b. Placenta increta:

- Placenta is attached to the myometrium.

b. Placenta percreta:

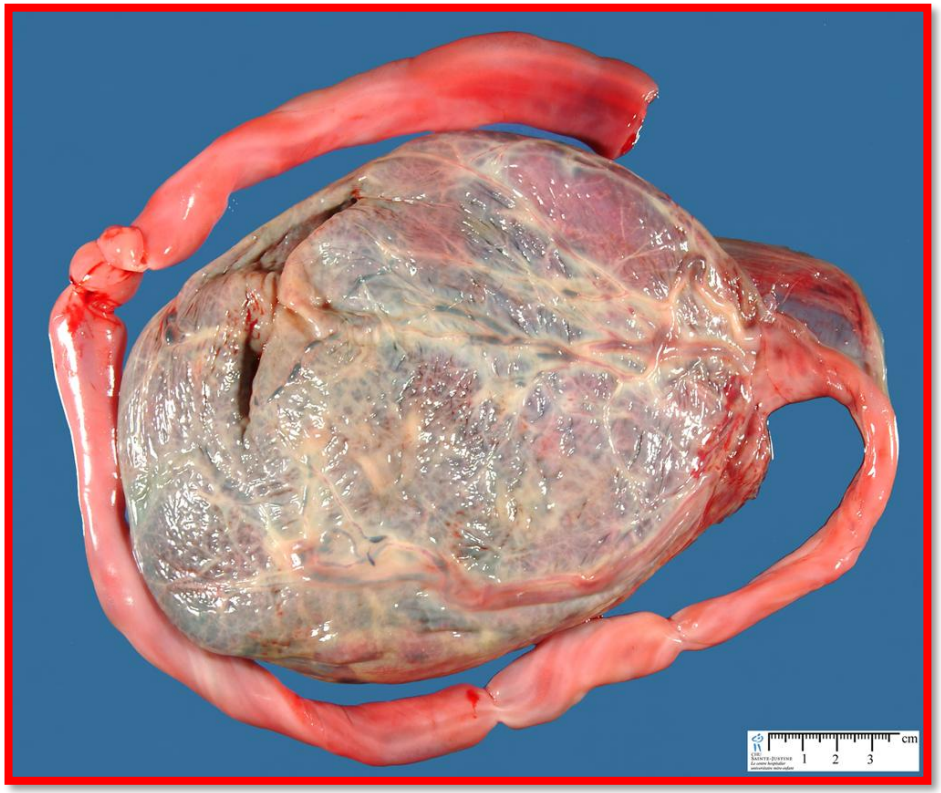
- The chorionic villi penetrate the myometrium all the way to the perimetrium (peritoneal covering of uterus).



- In these abnormalities the placenta fails to separate from the uterus after birth of the baby and may cause severe postpartum hemorrhage.

IV. According to the attachment of umbilical cord:

a. Marginal (Battledore) placenta:



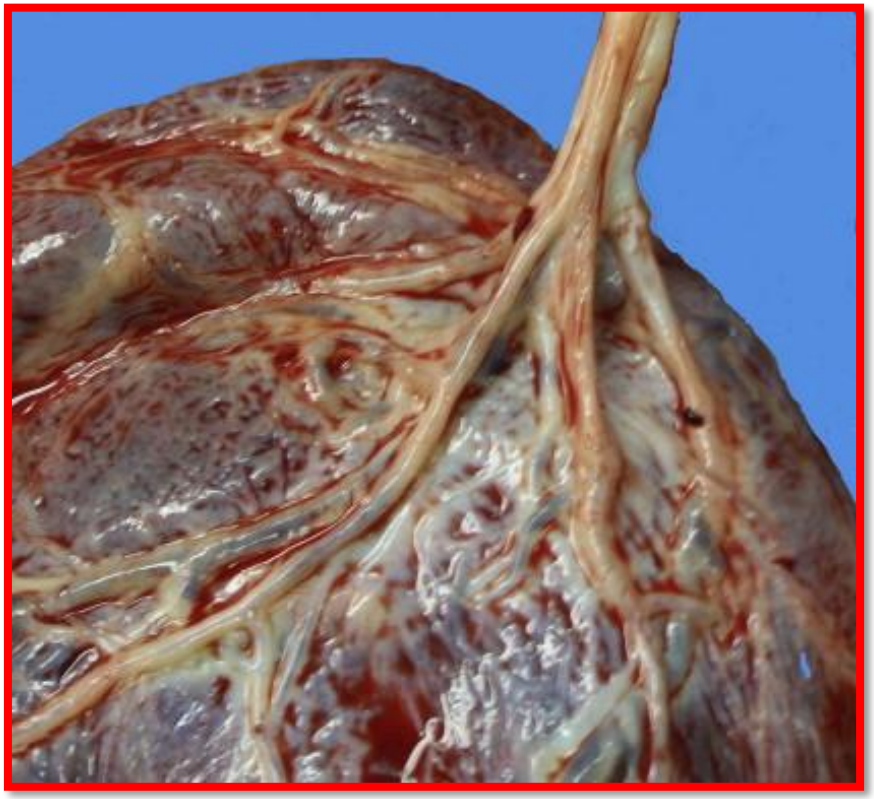


- **The cord is attached to the margin of placenta.**

- ***It is called so as its resemblance to the racquet used in badminton.***

b. Velamentous placenta:



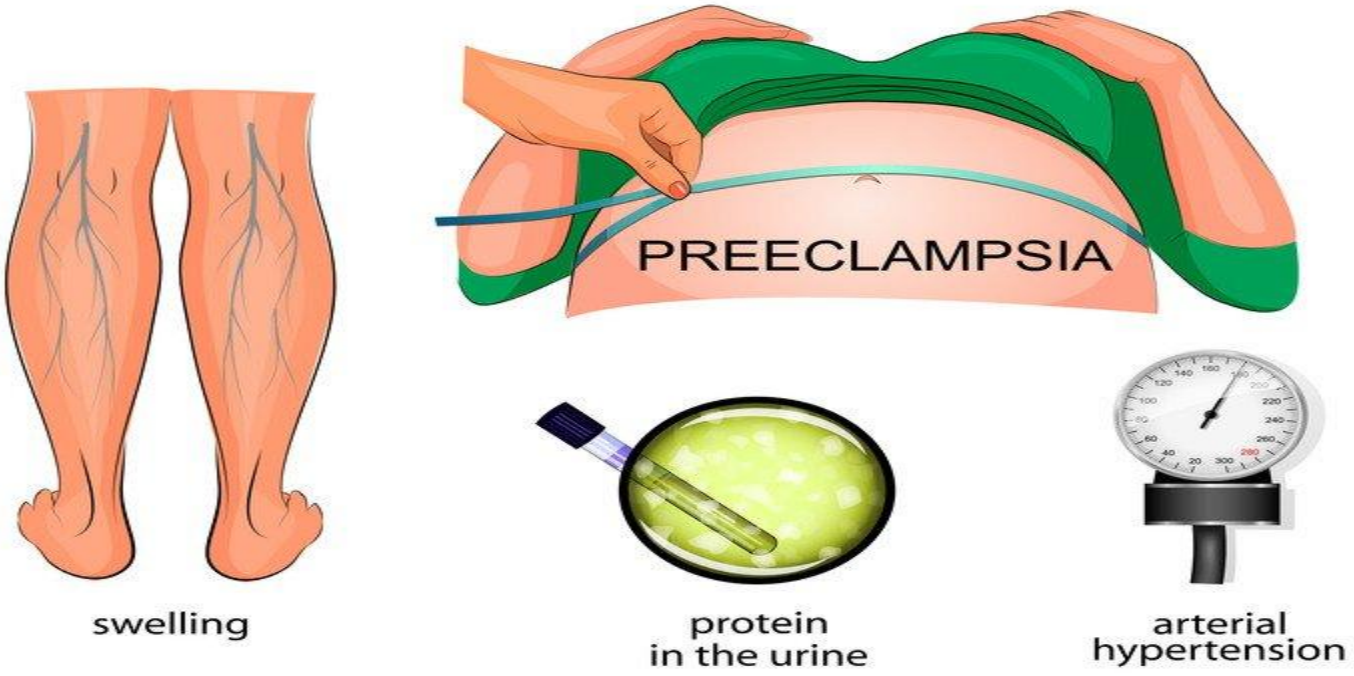


- The cord does not reach the placenta itself but is attached to amniotic membrane over the fetal surface of placenta.

- The umbilical vessels pass in the membrane to reach the placenta.

Preeclampsia

- **Maternal hypertension** in approximately 5% of pregnancies.
- The condition may progress to **eclampsia**, which is characterized by **seizures**.
- **IUGR**, **fetal death**, or **death of the mother**.
- Completely reversible by delivery of the baby.
- Risk factors include preeclampsia in a previous pregnancy, first pregnancy, obesity, family history of preeclampsia, multiple gestation (twins or more), hypertension and diabetes.



Umbilical cord

Characteristics of the normal umbilical cord at birth

- It is 30-80 cm long (average is 50 cm) and 2 cm in diameter.
- It is spirally twisted leading to false knots.
- It is covered by the amnion.



Umbilical cord

(1) The primitive umbilical cord is formed by the connecting stalk, vitelline duct and covered by the amniotic membrane.

(2) The connecting stalk contains the allantois and the umbilical vessels, consisting o f two arteries and one vein.

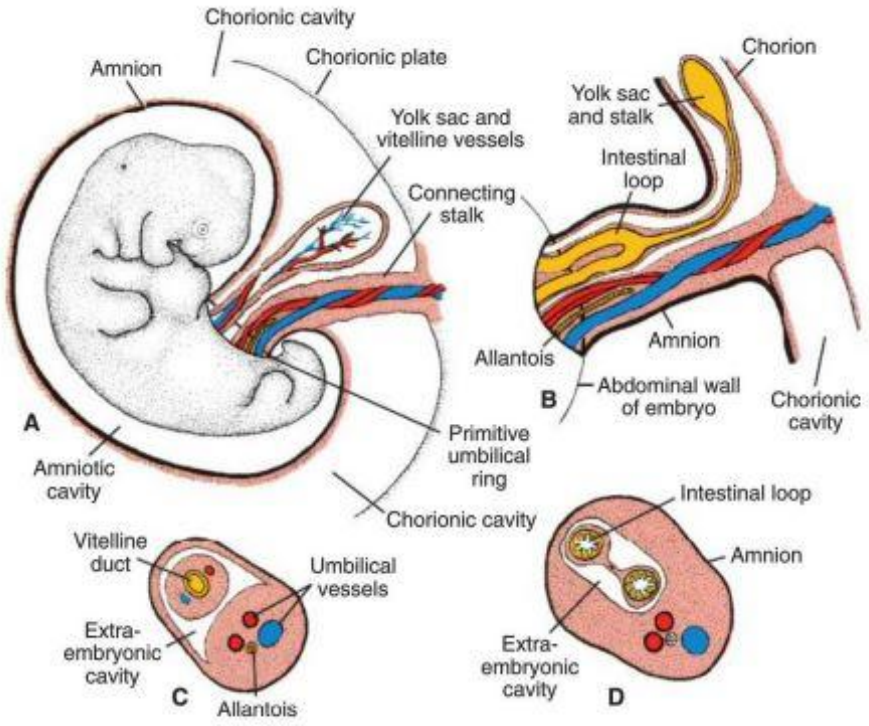
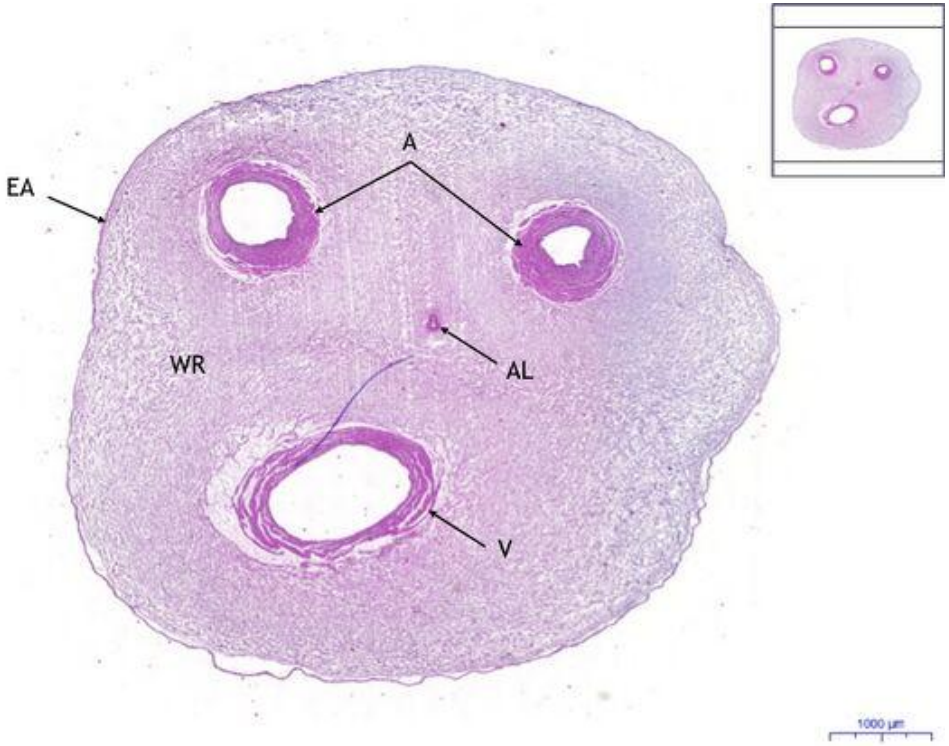


FIGURE 8.16 **A.** A 5-week embryo showing structures passing through the primitive umbilical ring. **B.** The primitive umbilical cord of a 10-week embryo. **C.** Transverse section through the structures at the level of the umbilical ring. **D.** Transverse section through the primitive umbilical cord showing intestinal loops protruding in the cord.

Umbilical cord

Contents at birth:

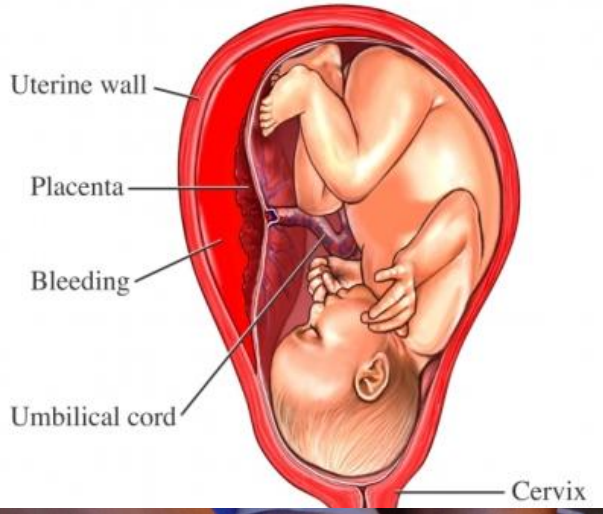
- Wharton's jelly.
- 2 Umbilical arteries.
- 1 Umbilical vein.
- Allantois (urachus).
- Remains of the vitelline duct.

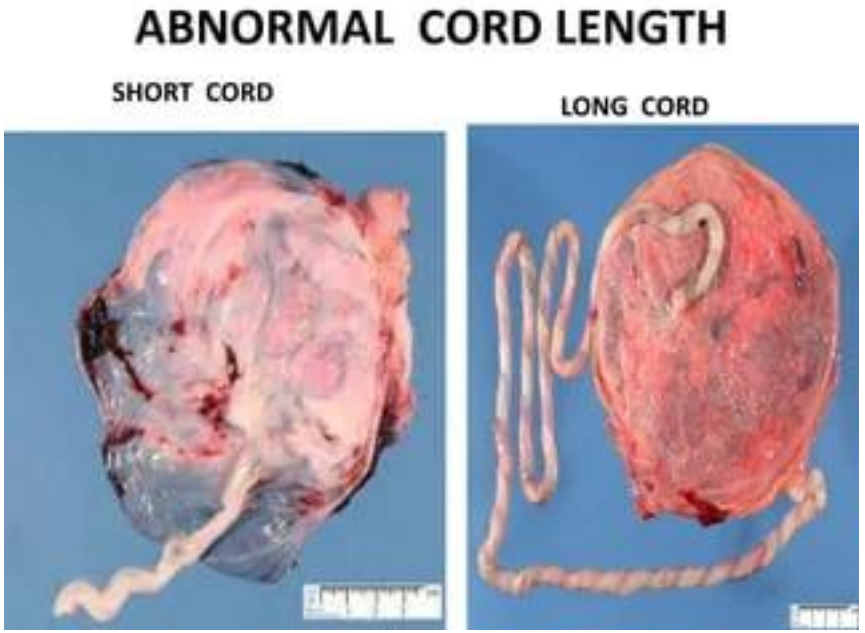


Umbilical cord

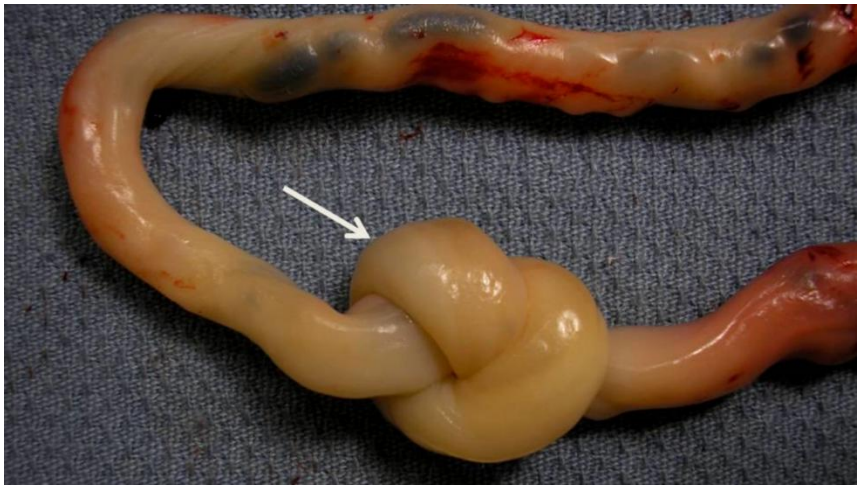
Congenital anomalies:

- Long cord: true knots, cord loop around neck.
- Short cord: abruptio placenta.

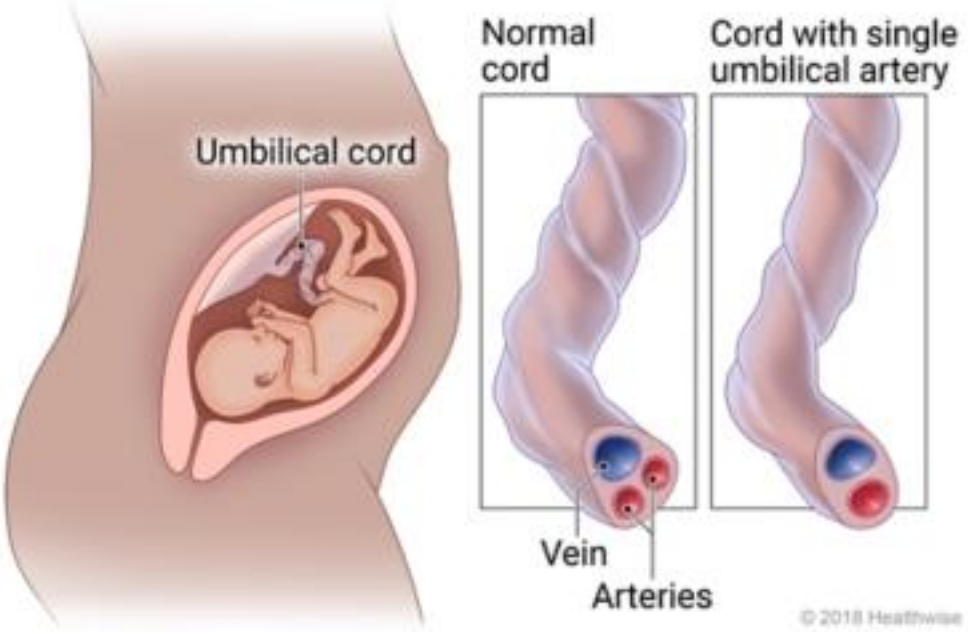








In 1 in 200 newborns, only a single umbilical artery is present, and these babies have approximately 20% chance of having cardiac, renal and other vascular defects.



Thank you

